CS177 Bioinformatics: Software Development and Applications

Lecture 21: AI for Drug Discovery and Design

Jie Zheng (郑杰)
PhD, Associate Professor
School of Information Science and Technology (SIST), ShanghaiTech University

Spring, 2024

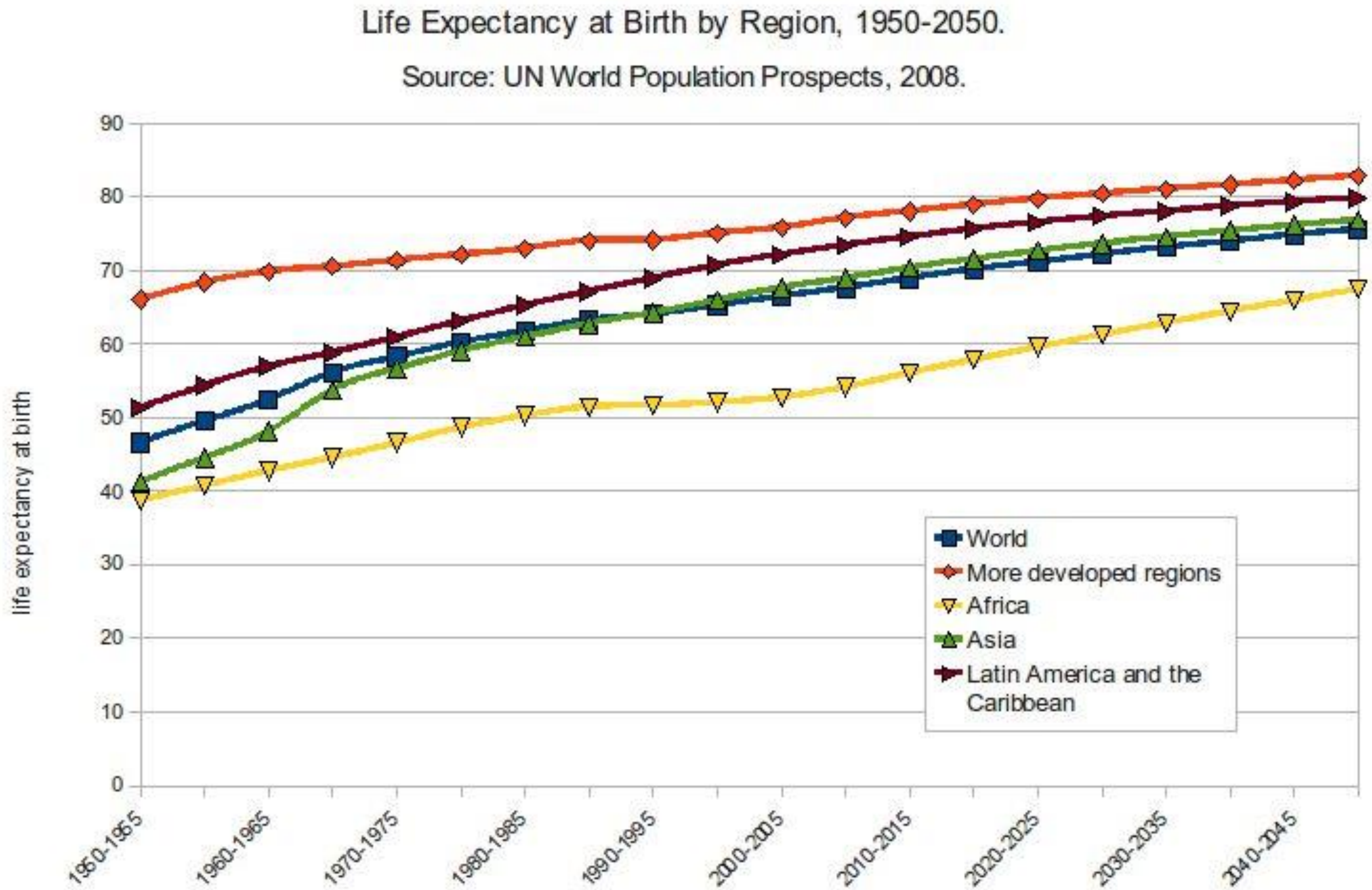
Credit: PPT slides courtesy from Prof. Fang Bai, SIAIS, SLST, SIST at ShanghaiTech

Outline

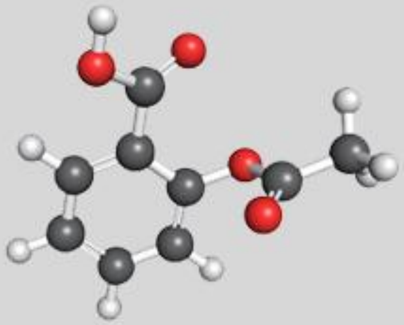
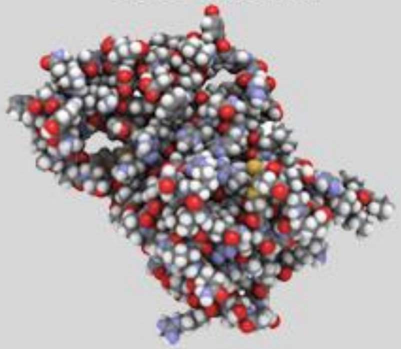
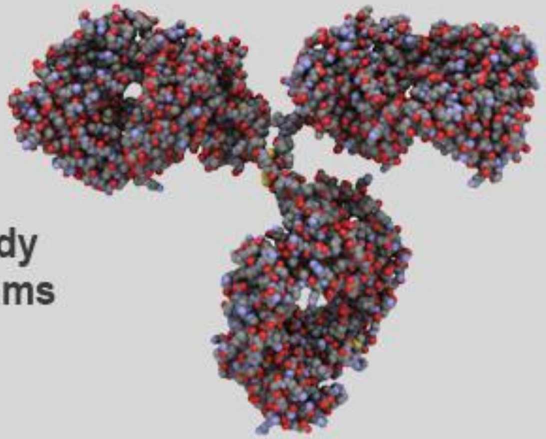
- Overview of drug discovery pipeline
- AI for drug development
- Drug target
- Drug action mechanisms
- Protein structure
- Drug-target interaction (DTI)
- ADMET (Absorption, Distribution, Metabolism, Excretion, Toxicity)

Overview of Drug Discovery Pipeline

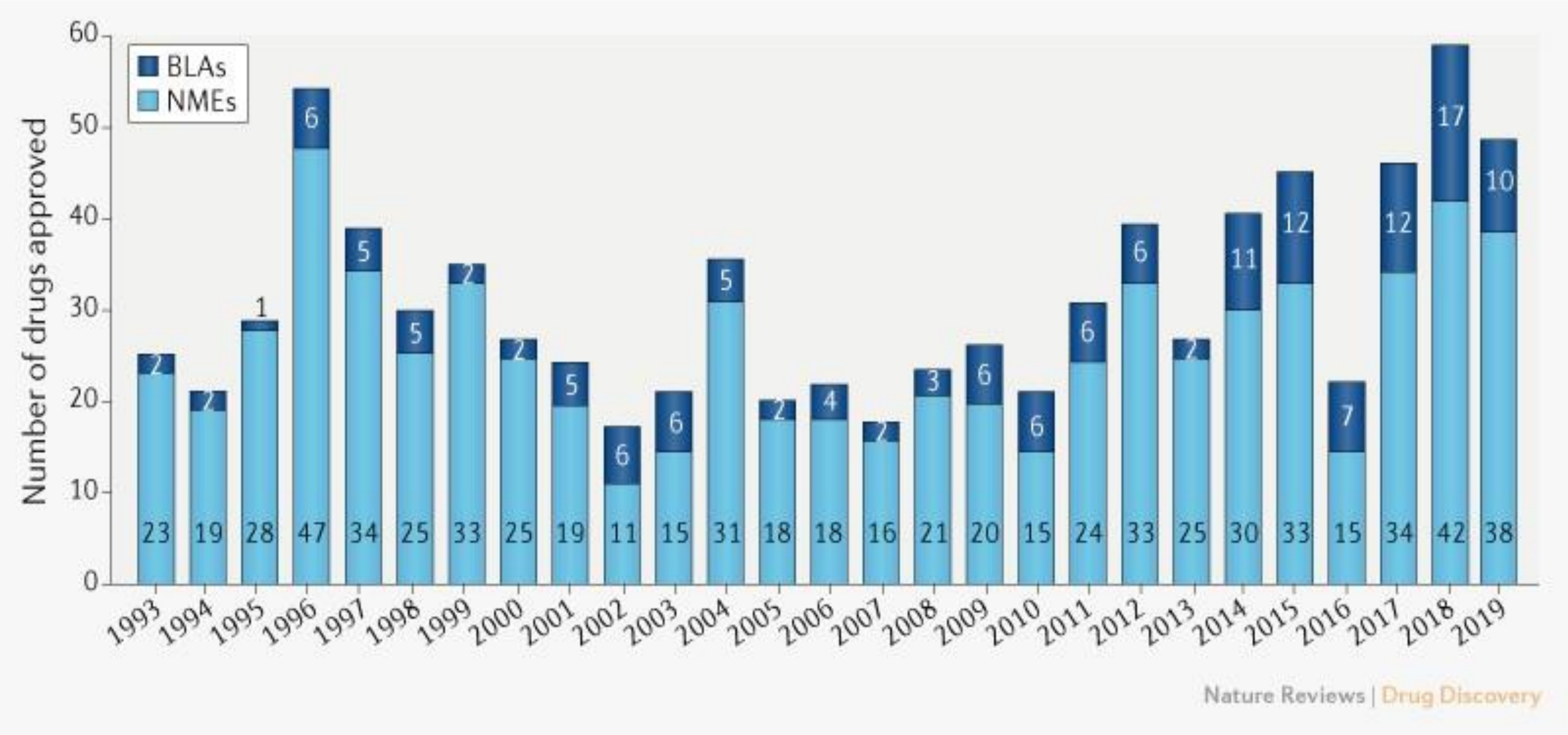
Driving force for drug development



Size & complexity of small molecular drugs / biological medicine

	Small Molecule Drug	Large Molecule Drug	Large Biologic
Size	Aspirin 21 atoms 	hGH ~ 3000 atoms 	 IgG Antibody ~ 25,000 atoms
Complexity	Bike ~ 20 lbs 	Car ~ 3000 lbs 	 Business Jet ~ 30,000 lbs (without fuel)
	阿司匹林	人体生长激素	免疫球蛋白

Novel FDA approvals since 1993

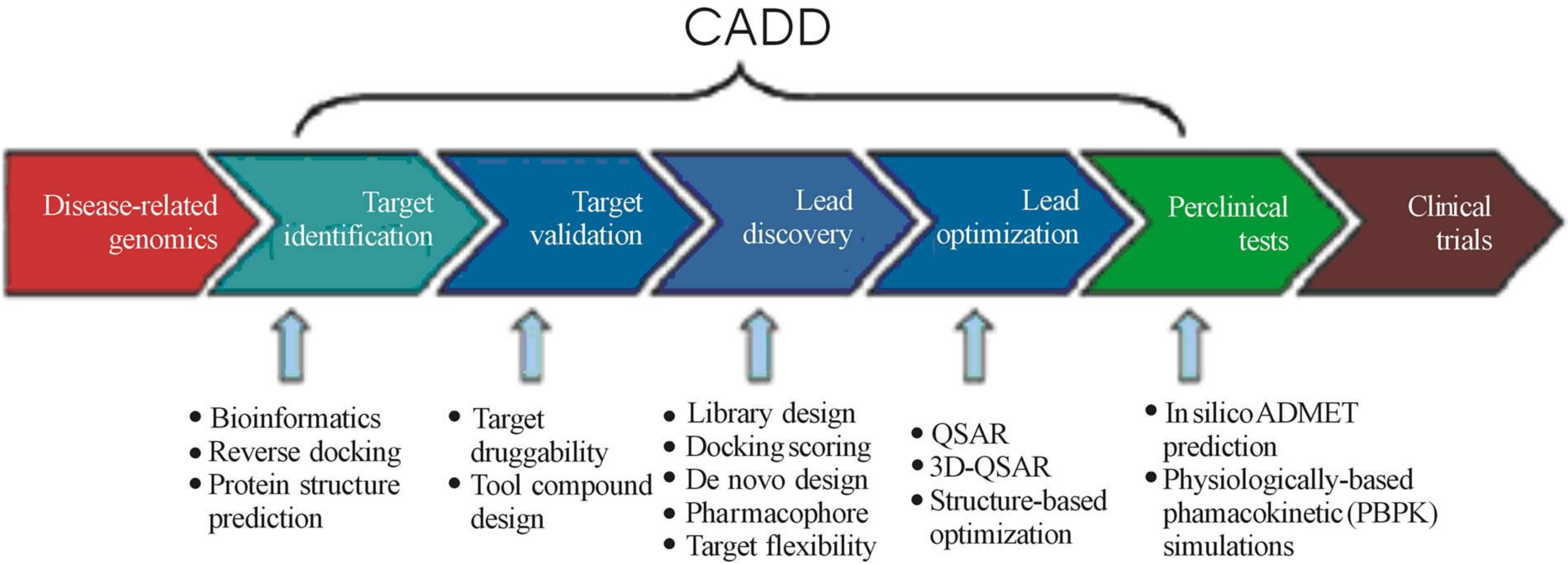


Annual numbers of new molecular entities (NMEs) and biologics license applications (BLAs) approved by the FDA' s Center for Drug Evaluation and Research (CDER).

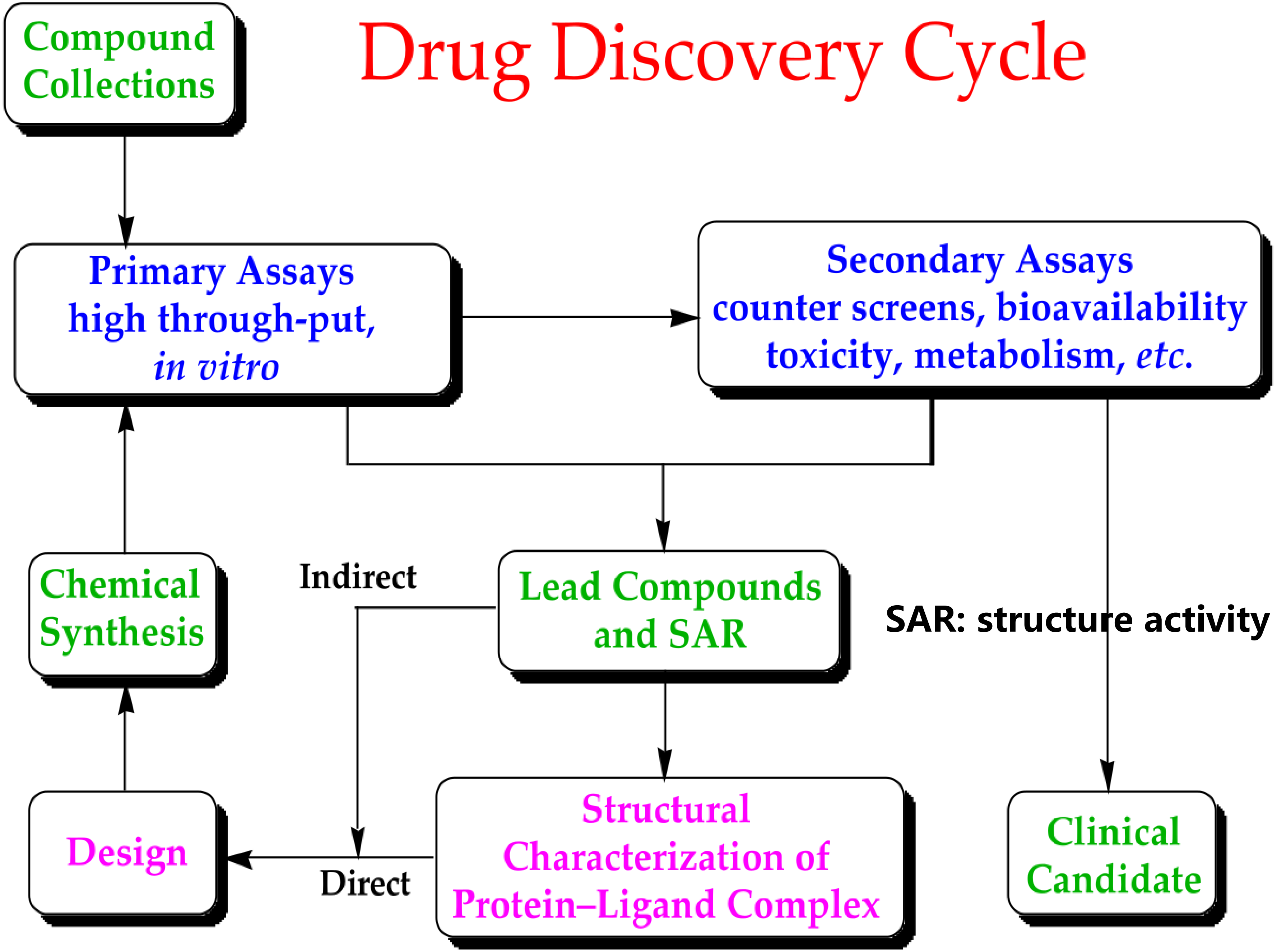
NMEs: Include all peptides and oligonucleotides of up to 40 amino acids in length.
BLAs: Protein-based candidates

Nature Reviews Drug Discovery, 2020, 19, 79

Computational Aided Drug Design (CADD)
Artificial Intelligence Drug Design (AIDD)



Drug Discovery Cycle



SAR: structure activity relationship

AI for Drug Development

Lots of things happening: time for critical evaluation of where we are

nature

SPOTLIGHT | 30 May 2018

How artificial intelligence is changing drug discovery

Machine learning and other technologies are expected to make the hunt for new pharmaceuticals quicker, cheaper and more effective.

Pfizer

SANOFI

Roche

IBM

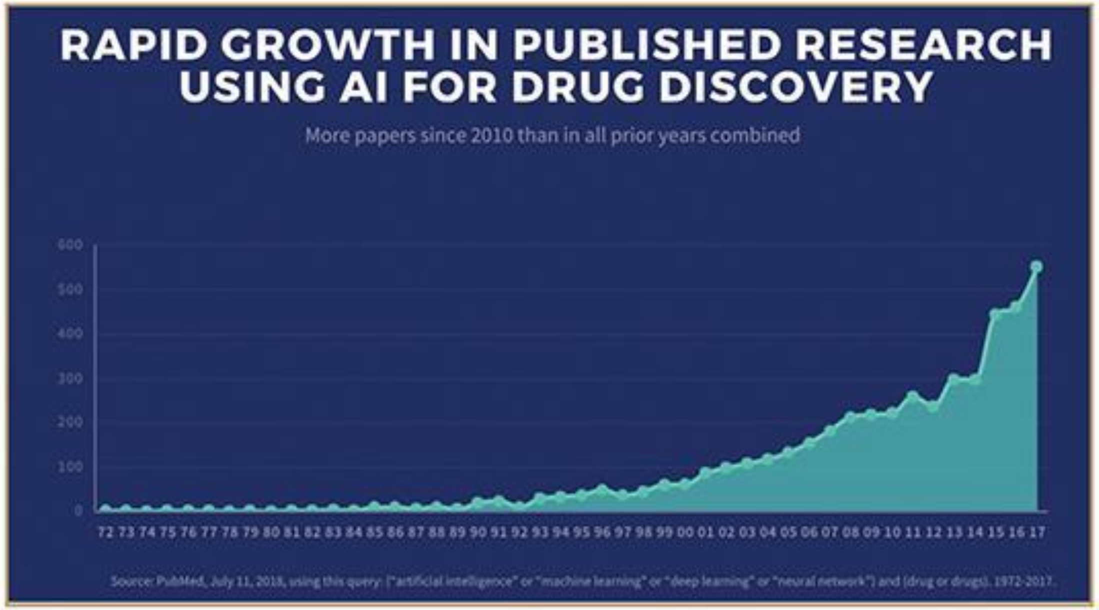
Exscientia

Genentech
IN BUSINESS FOR LIFE

immuno-oncology drugs

metabolic-disease therapies

cancer treatments



Here come robots…

12 June 2007

"Adam" to **carry out the entire scientific process on its own**: formulating hypotheses, designing and running experiments, analyzing data, and deciding which experiments to run next.

Adam

- Identify the function of a gene
- Generate hypothesis which genes are key enzyme
- Validate by itself (9/10 are right)

Eva

- Test compound (4/100 are active)
- Discover Compounds against malaria parasites

Drug Discovery and Development Timeline

Mfg.: Manufacturing

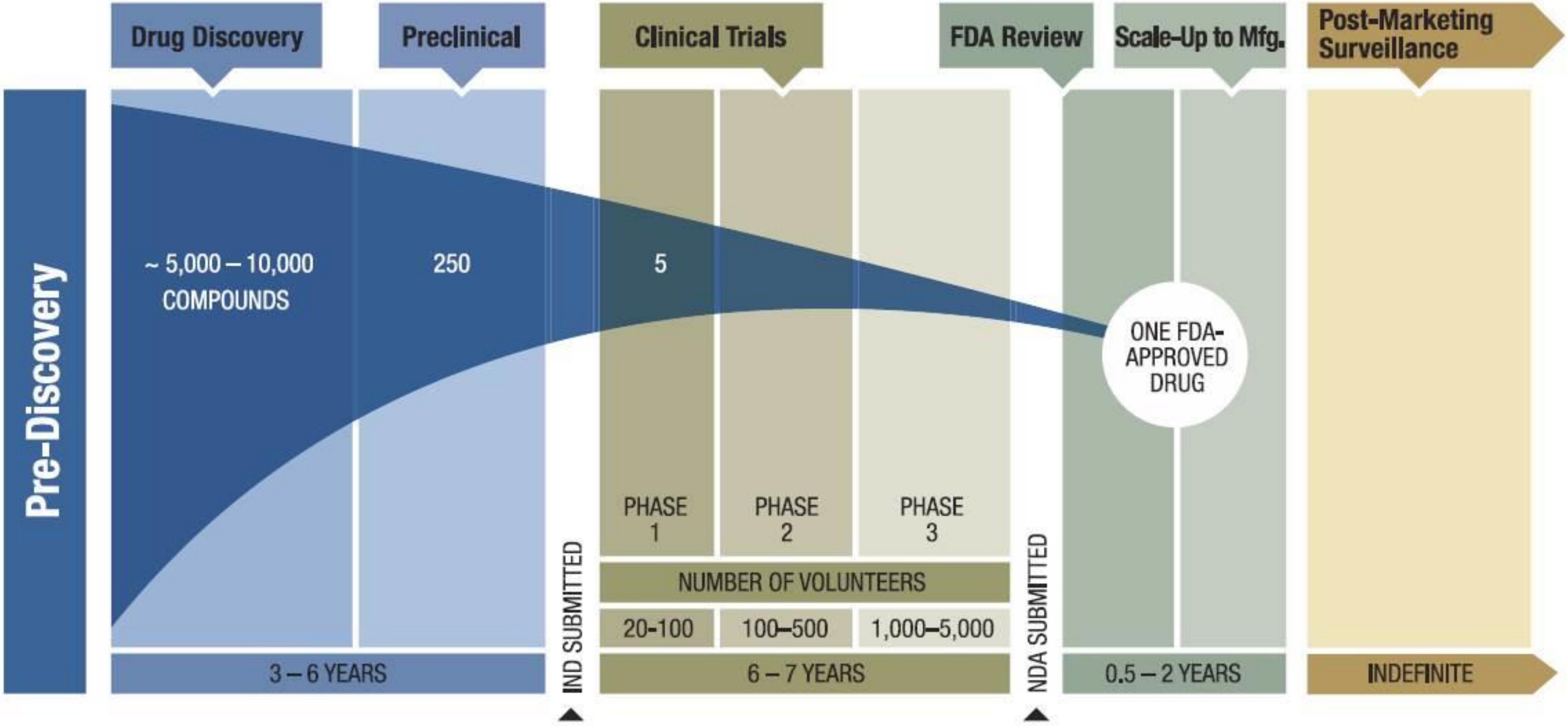


Figure adapted from: https://friendsofcancerresearch.org/cancer-drug-approval

Our AI system has revealed a novel wide-indication target, and a corresponding drug candidate in under 18 months and at roughly 1/10th of the typical cost associated with similar programs.

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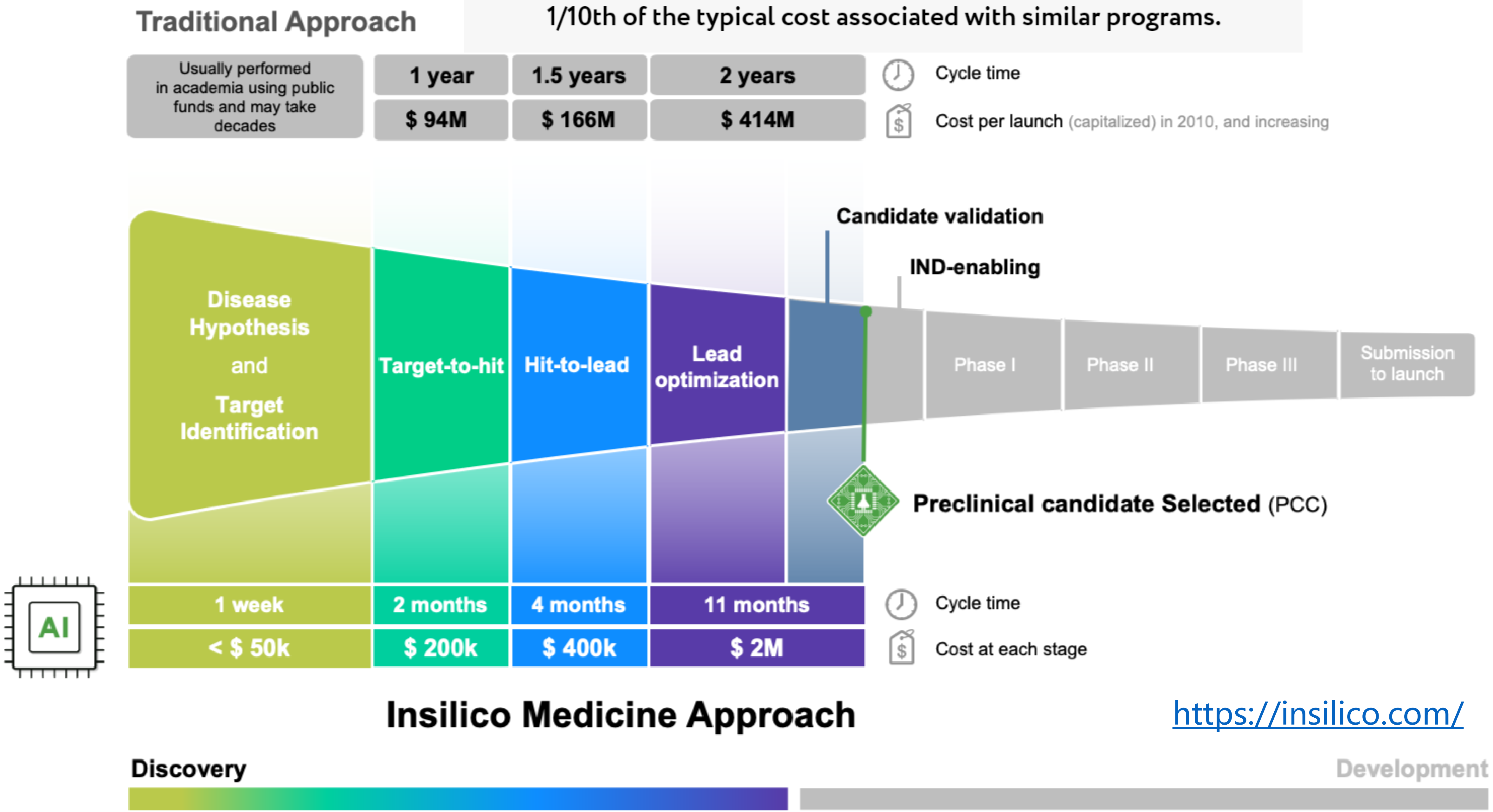
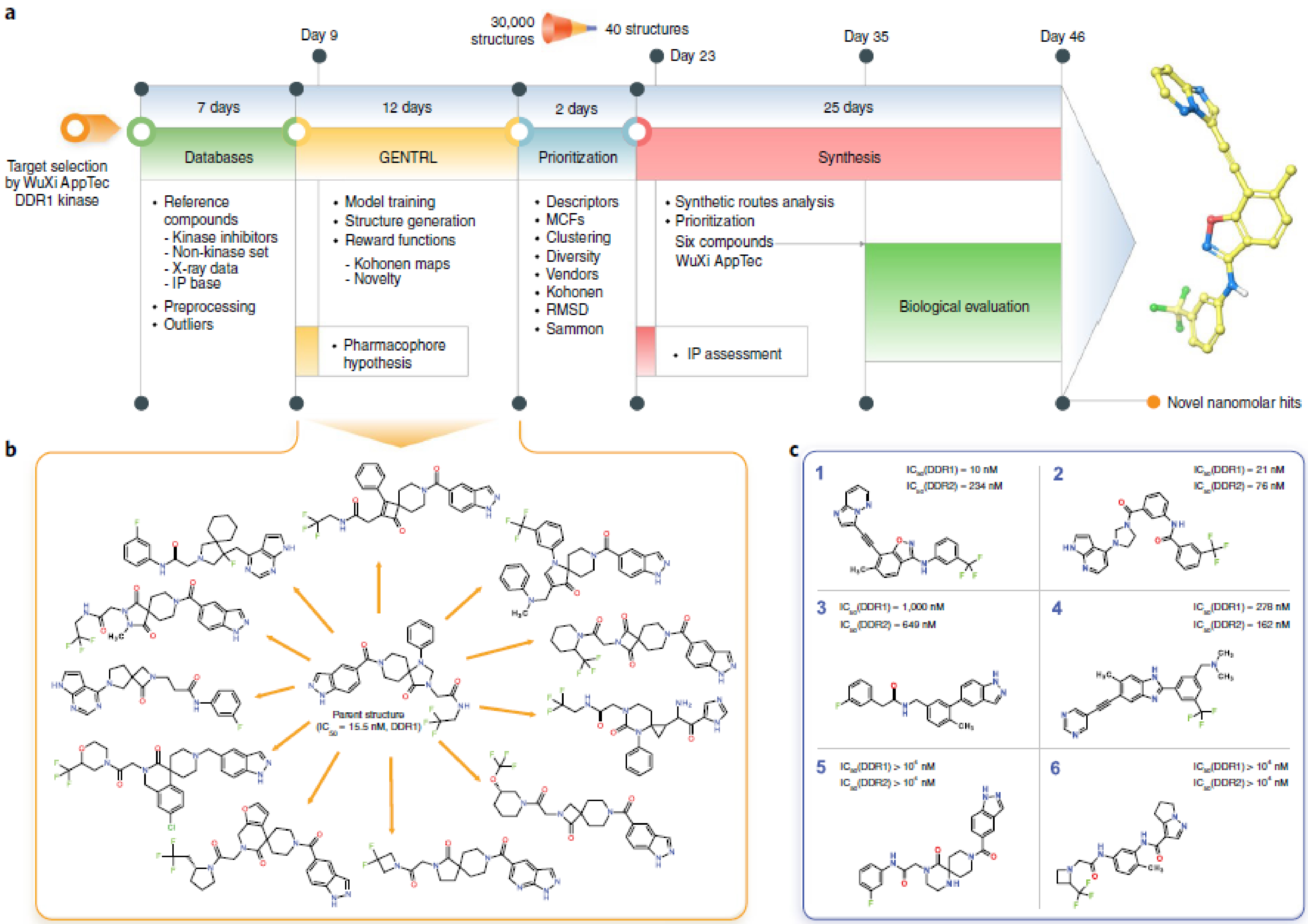


https://insilico.com/

Generative Tensorial Reinforcement Learning (GENTRL) model

Deep learning enables rapid identification of potent DDR1 kinase inhibitors



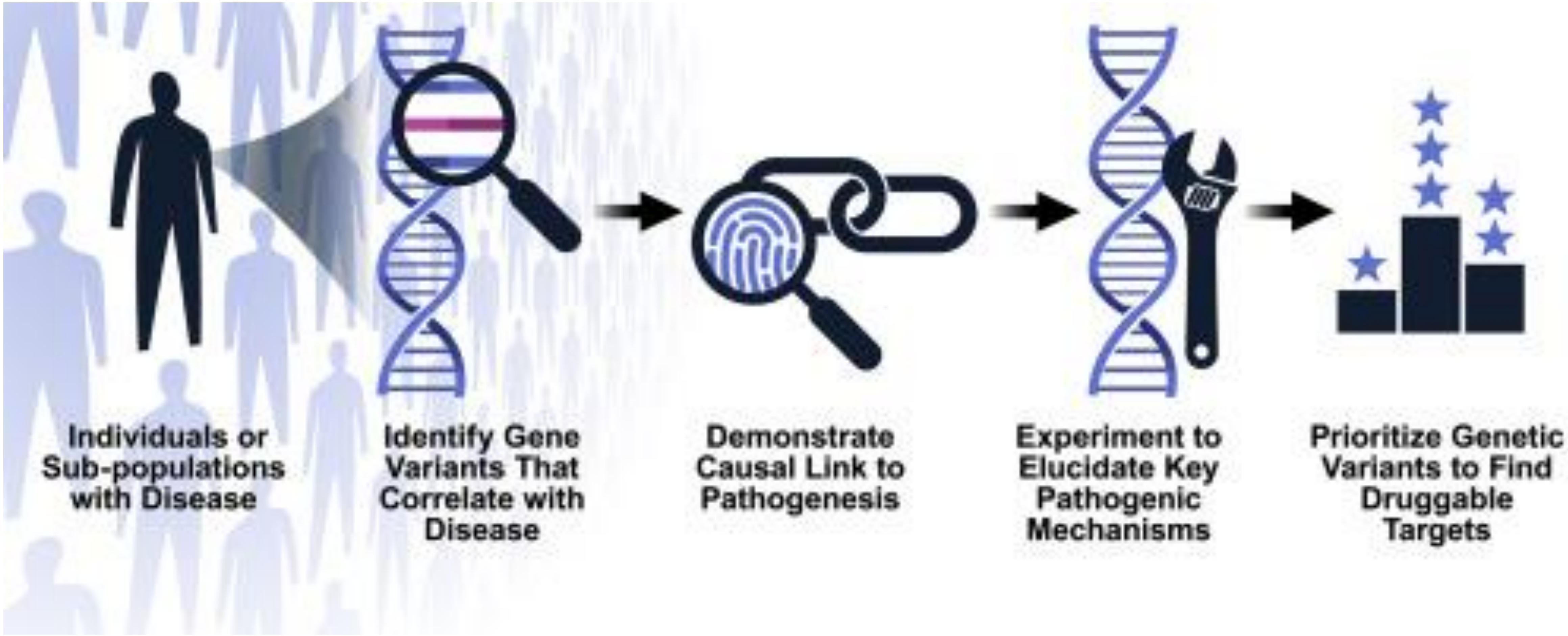
Nature Biotechnology, 2019,1038–1040

AI can Help in Many other areas of Drug Discovery

https://insilico.com/

Drug Target

How to define a potential drug target?



How to design drugs against a target?

Traditional Chinese Medicine

Modern Drug Development

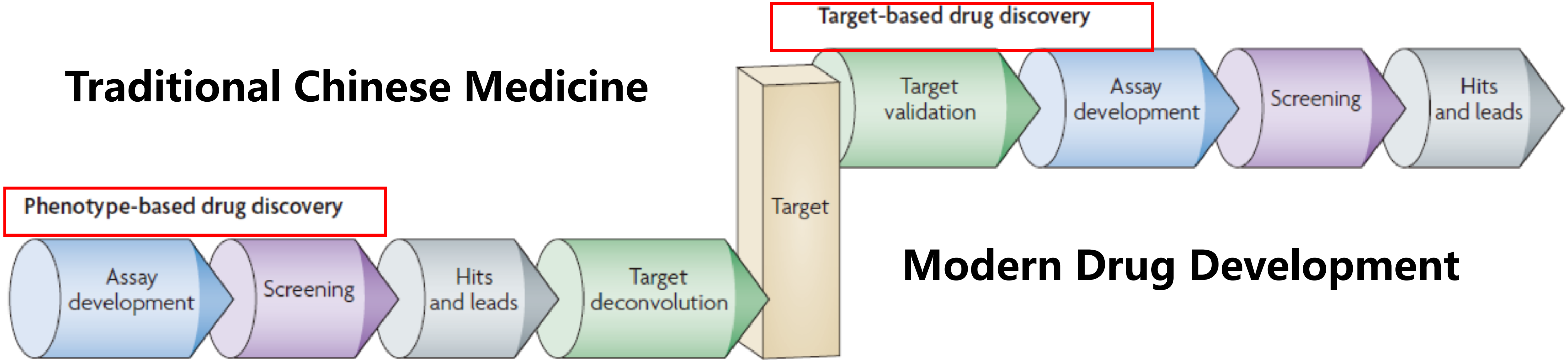


Figure 1 | **Phenotype-based versus target-based drug discovery.** The diagram illustrates the early phase of drug discovery, in which the aim is to identify target and lead molecules. In the phenotype-based approach, lead molecules are obtained first, followed by target deconvolution to identify the molecular targets that underlie the observed phenotypic effects. In the target-based approach, molecular targets are identified and validated before lead discovery starts; assays and screens are then used to find a lead.

Automated discovery of perturbed genes using expression data

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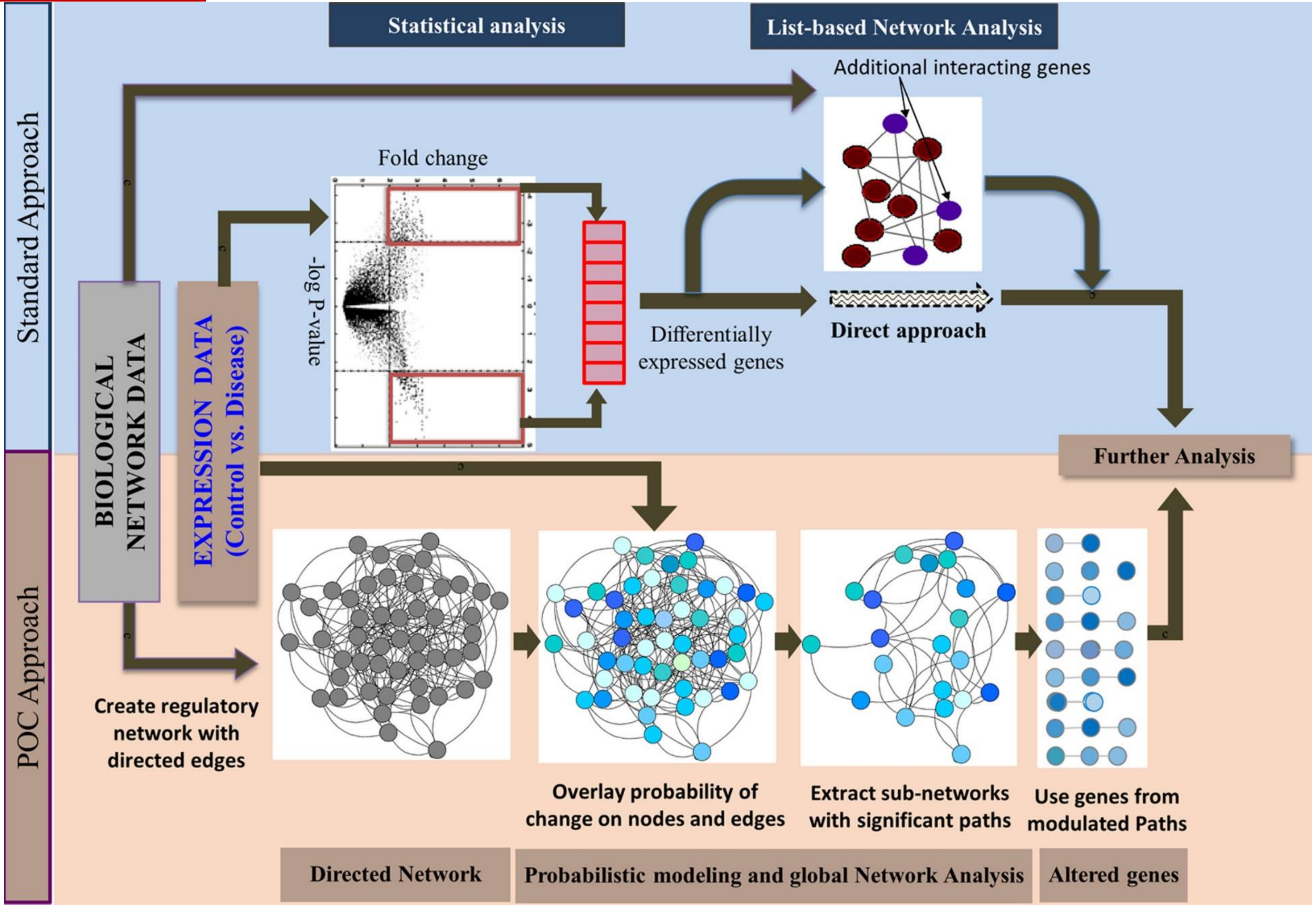


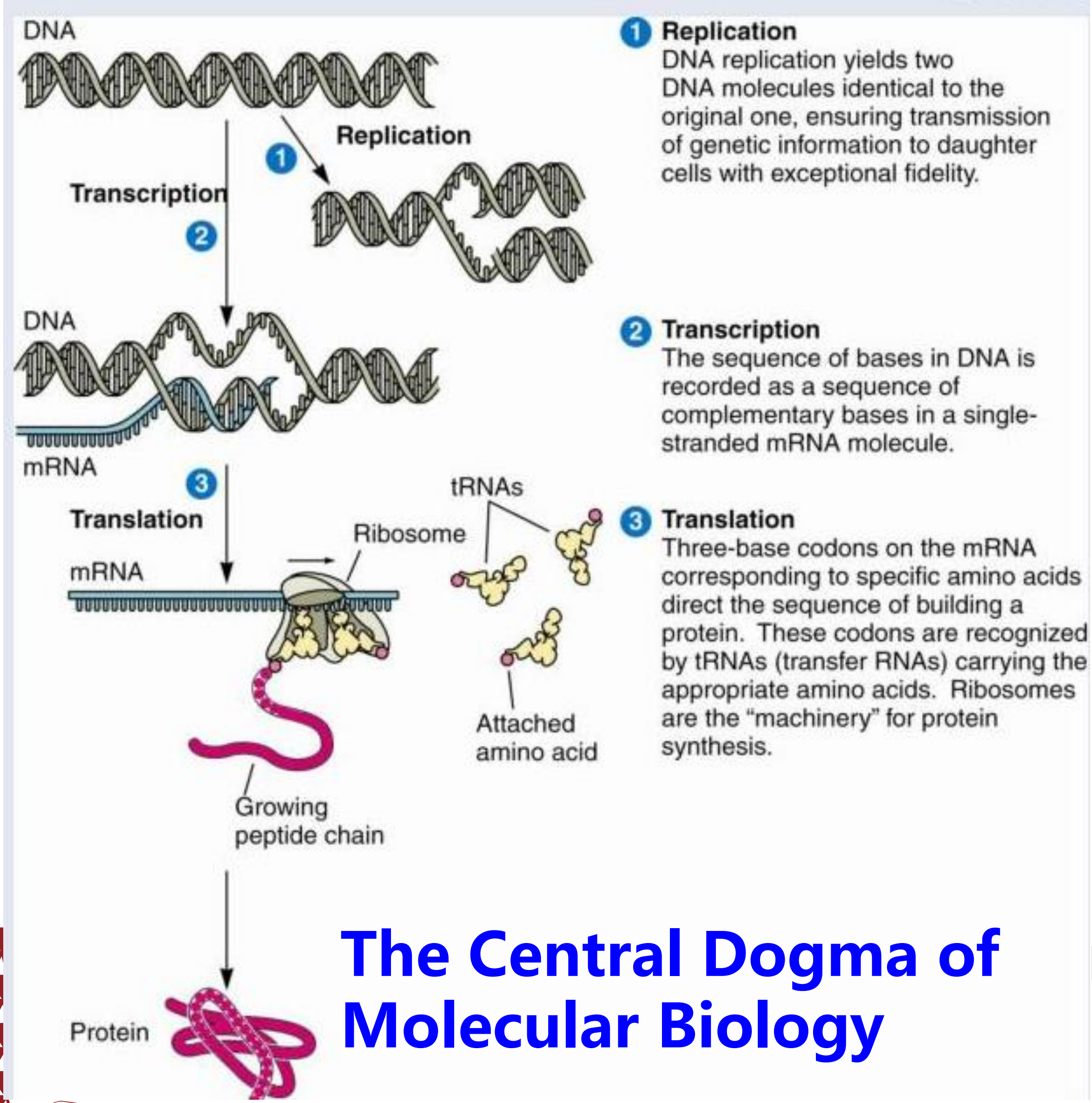
POC: Probability of Change

Comput. Biol. Med 2015, 67:29-40

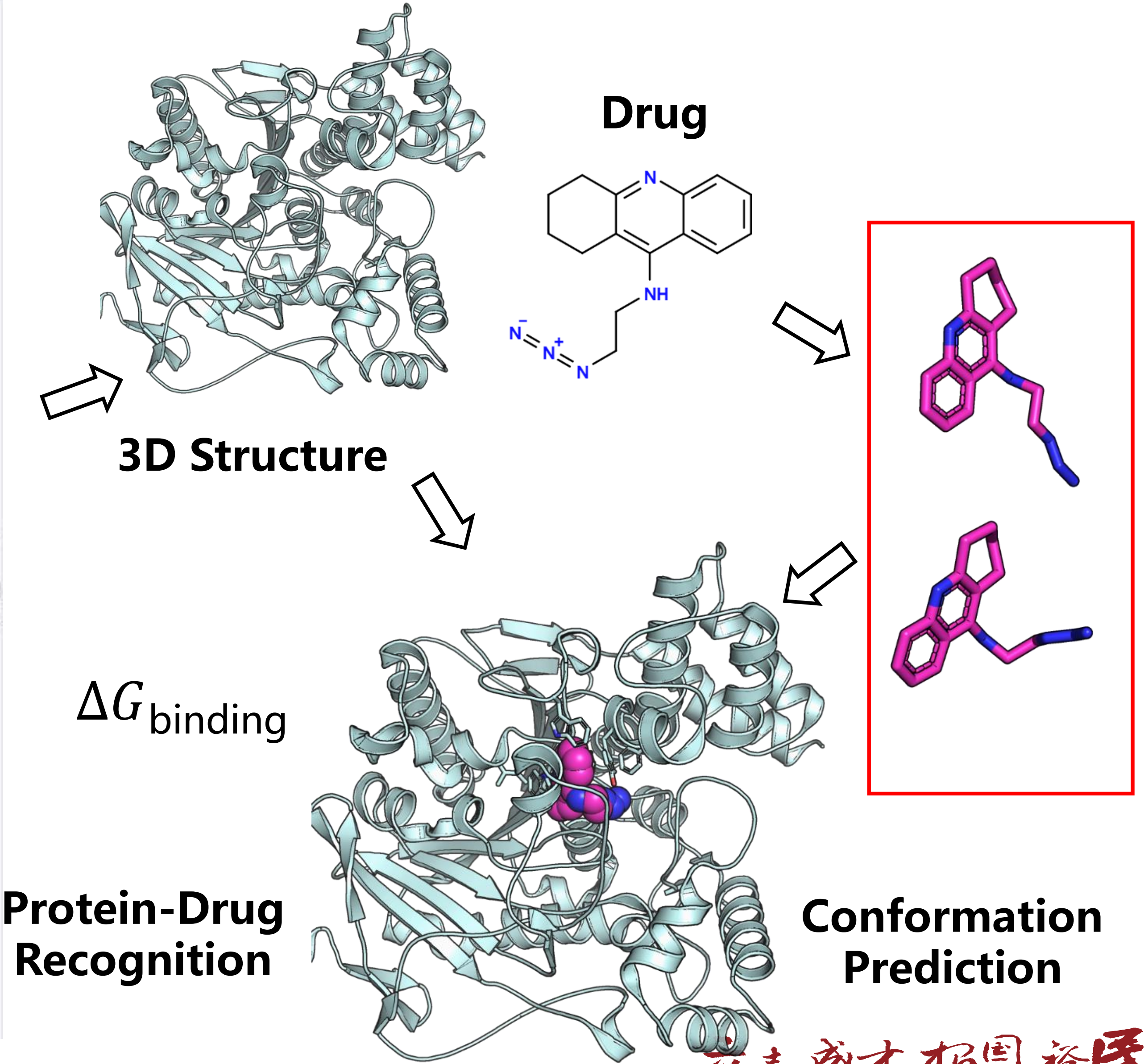
立志成才报国裕民

Drug Action Mechanisms

Central dogma of molecular biology vs drug design

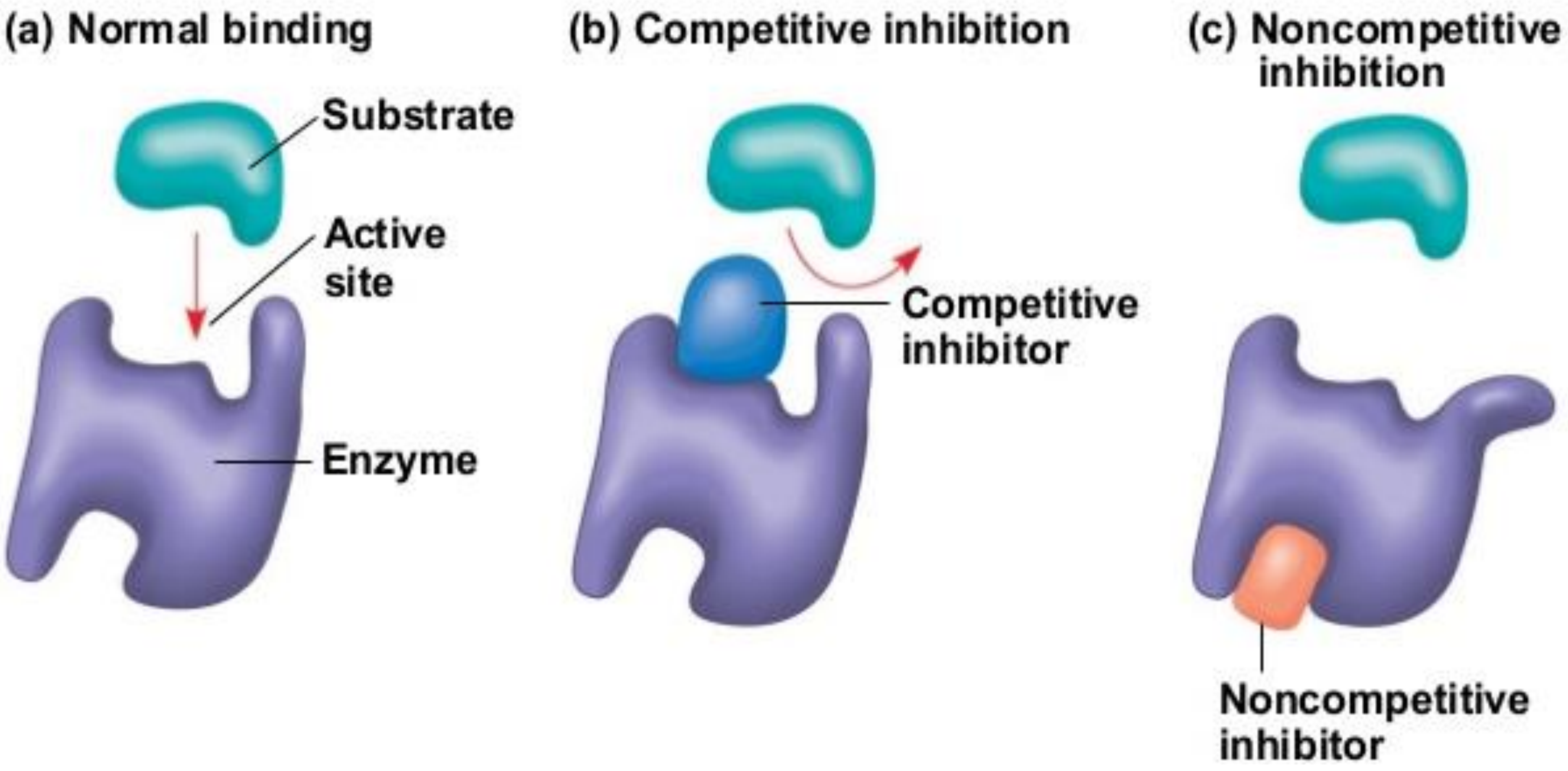


The Central Dogma of Molecular Biology



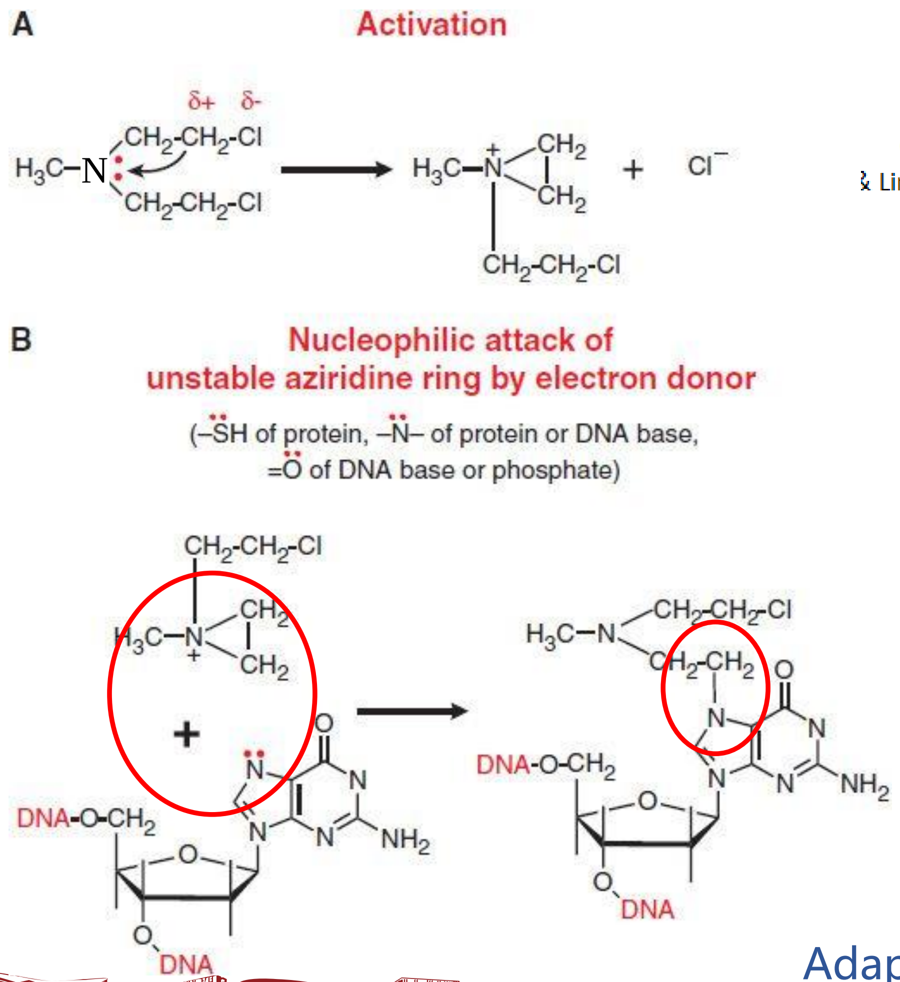
Drug action mechanisms

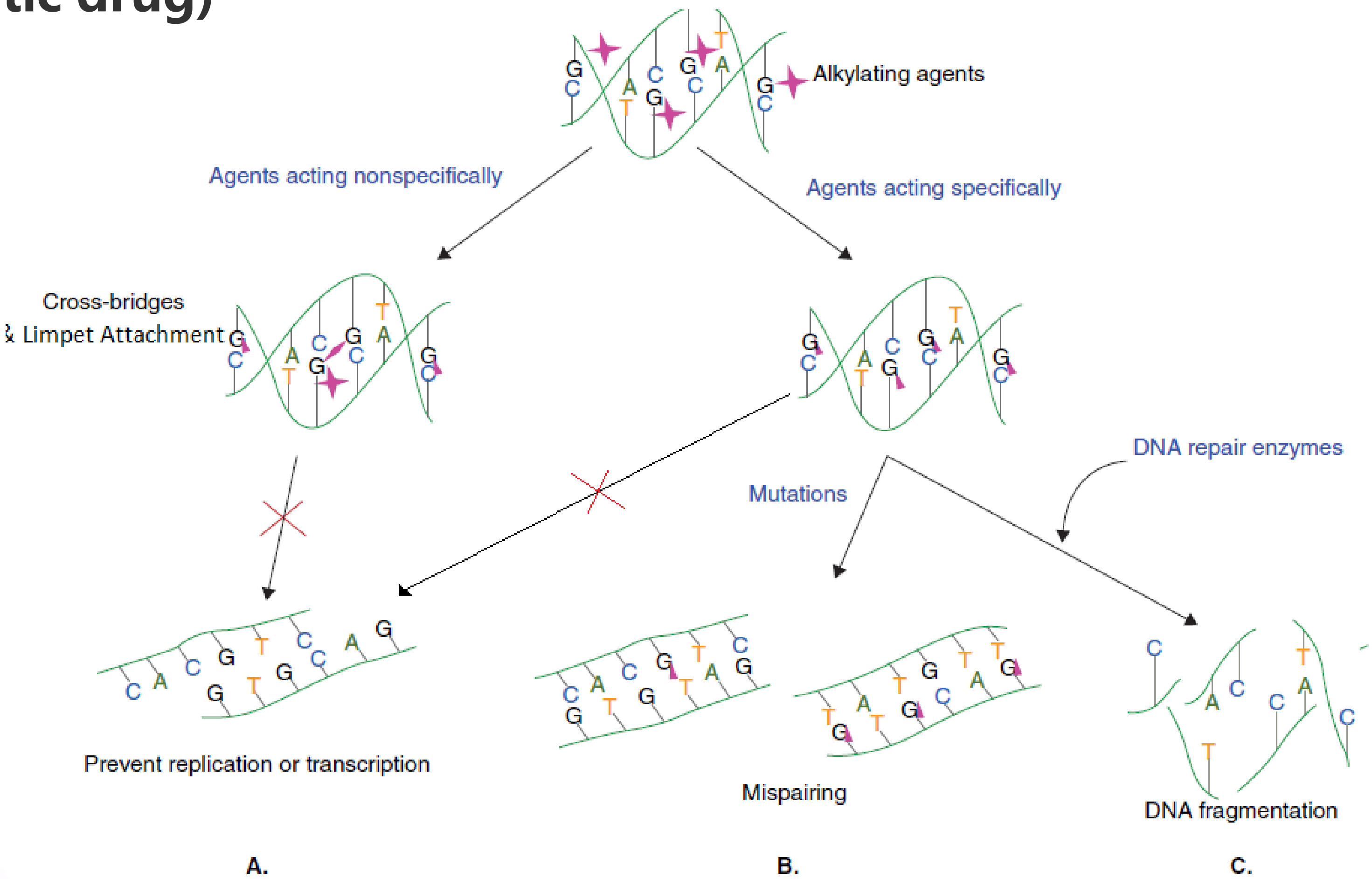
Enzyme Inhibitors



Drug action mechanisms

Cytotoxic drug (chemotherapeutic drug)
细胞毒性药物：e.g. 烷化剂



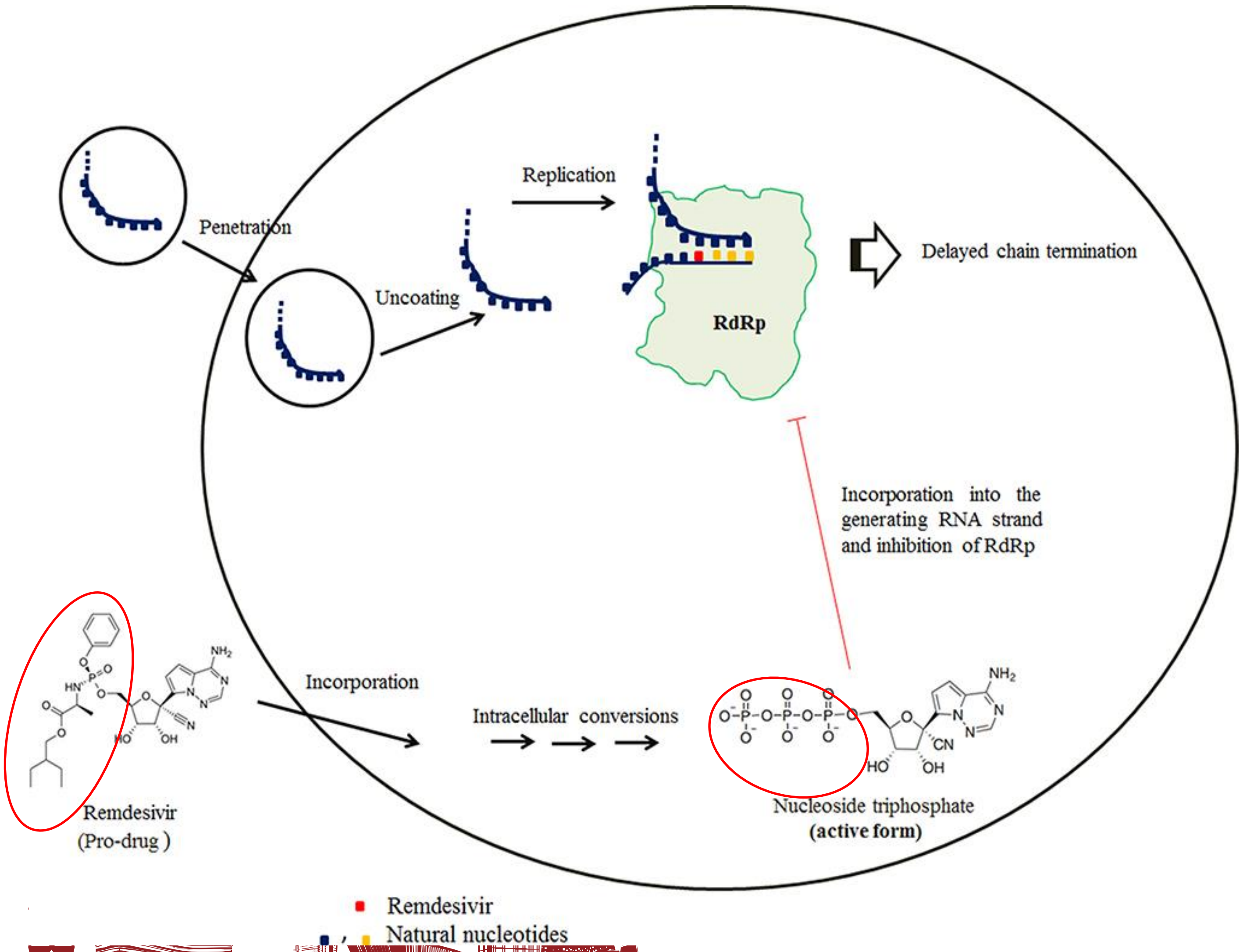


A.	B.	C.
阻碍DNA链解旋	引起碱基对错配，从而诱发突变	过度激活DNA修复酶，导致DNA断裂

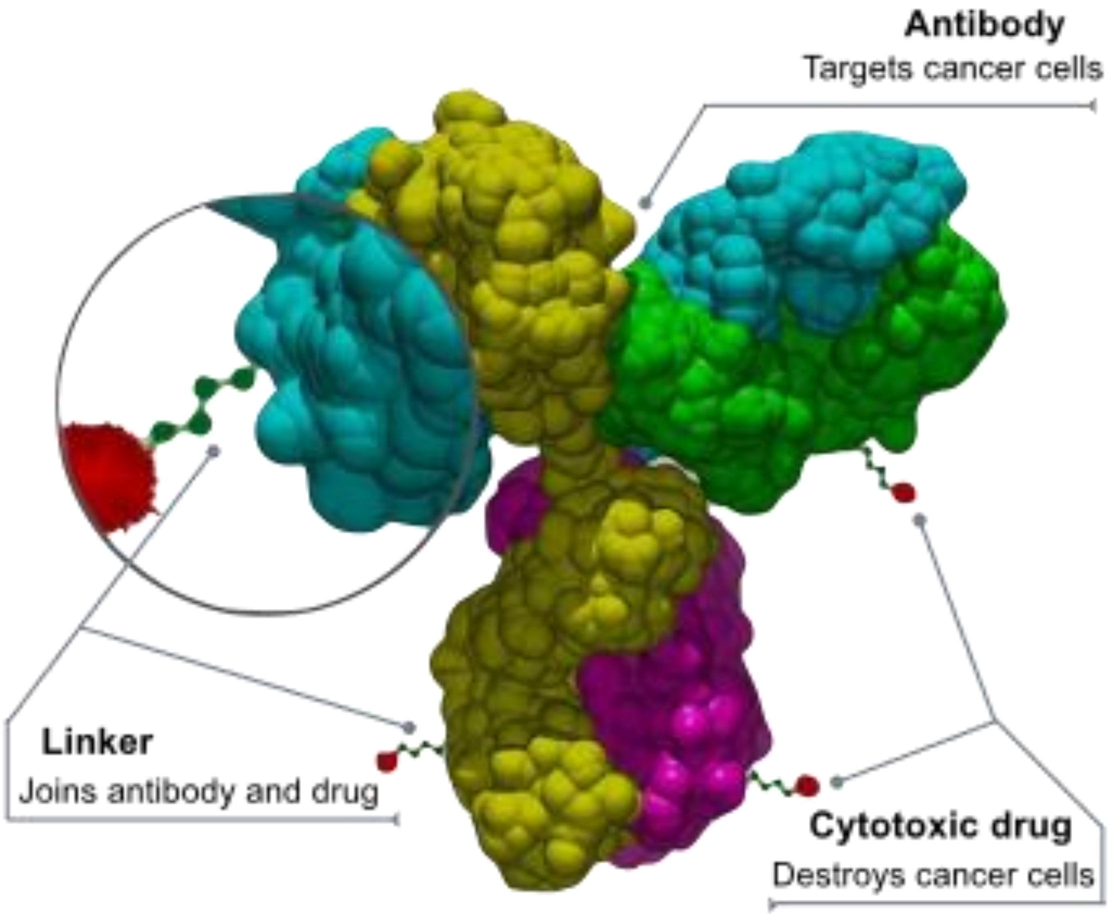
Adapted from Ralhan and Kaur, 2007

Drug action mechanisms

RNA drug Remdesivir

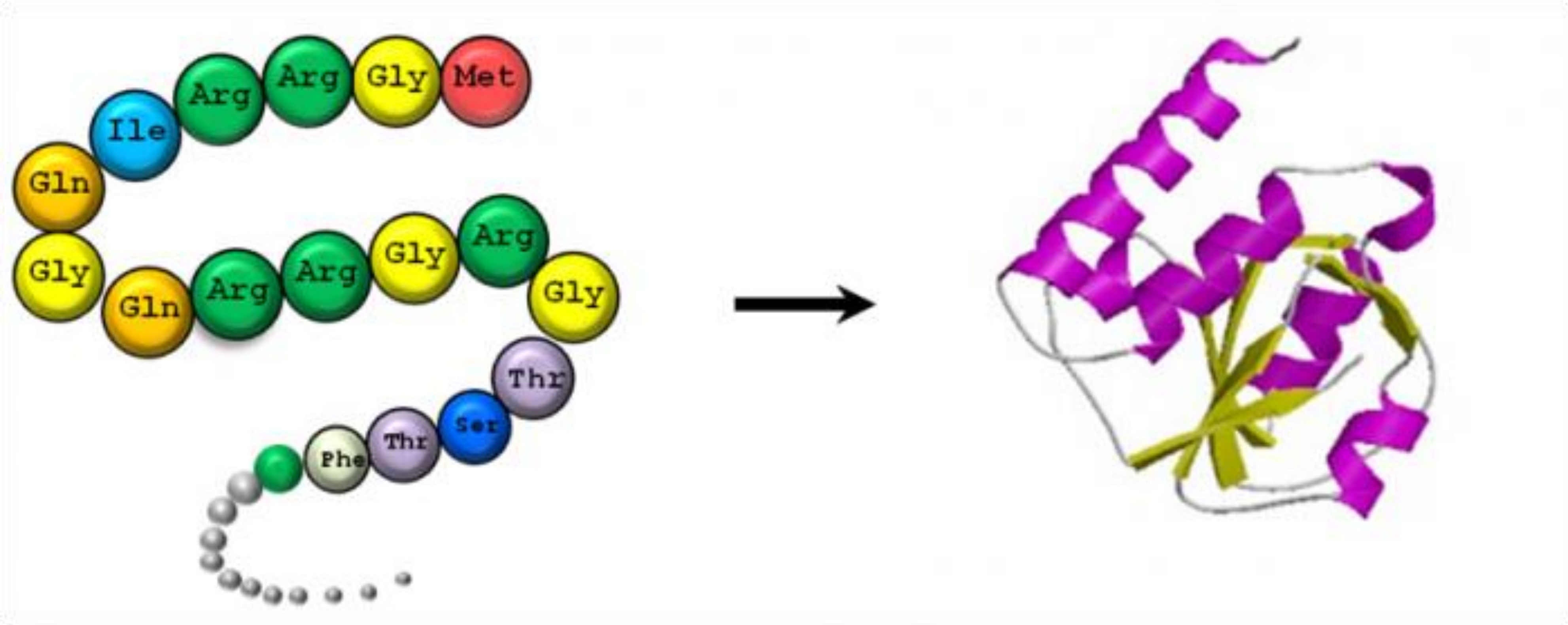


Antibody-drug conjugate

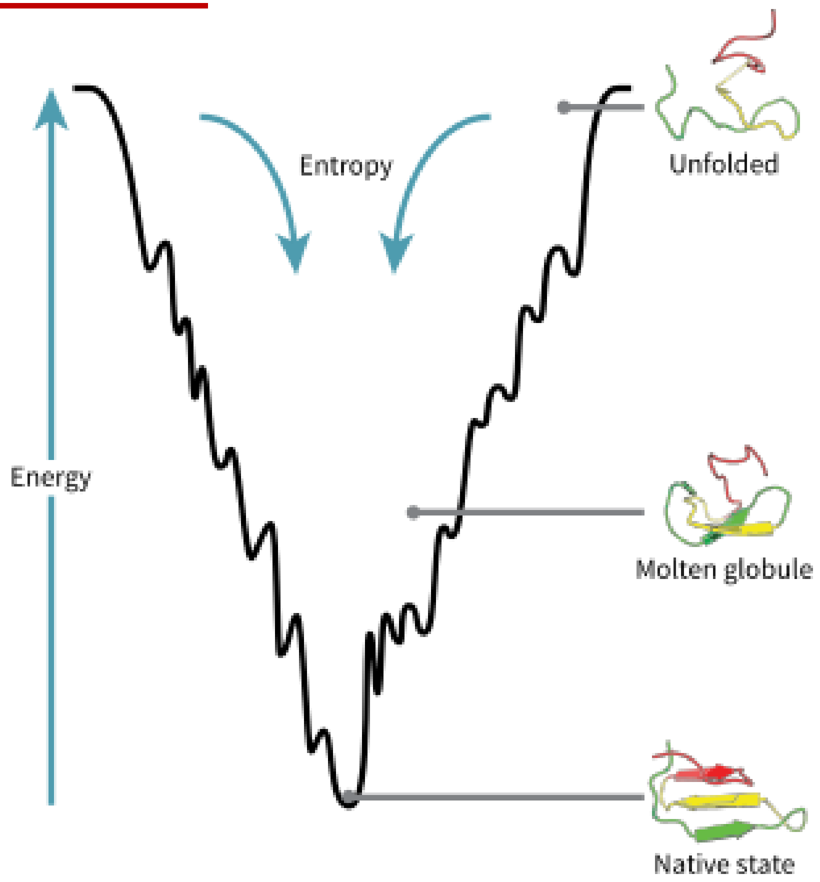


Protein Structure

Protein structure prediction

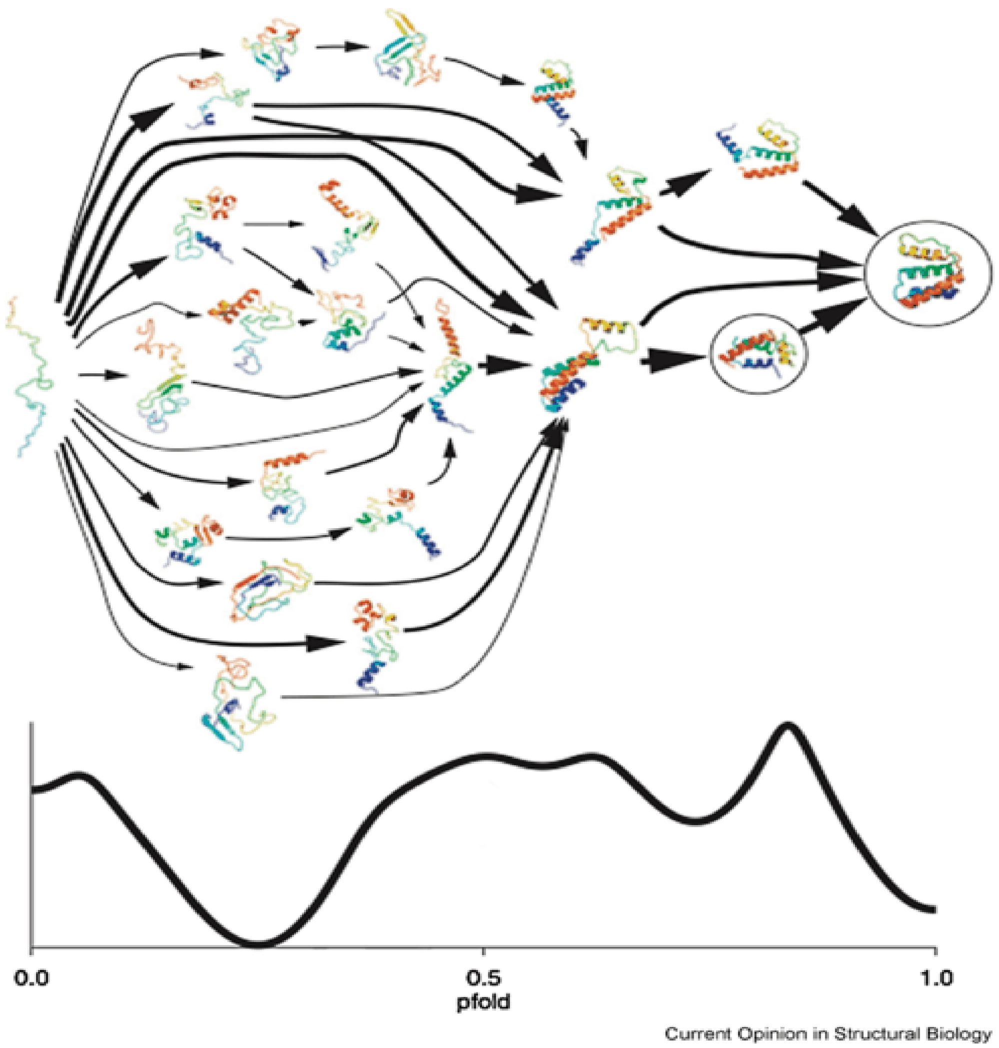


Protein folding: energy landscape and folding funnel



$$\Delta G = \Delta H^{\circ}_{sys} - T\Delta S_{system}$$

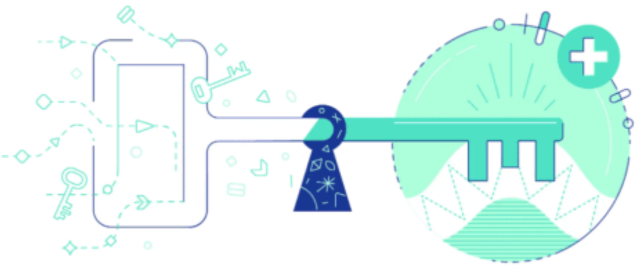
Enthalpy (焓)　　Entropy (熵)

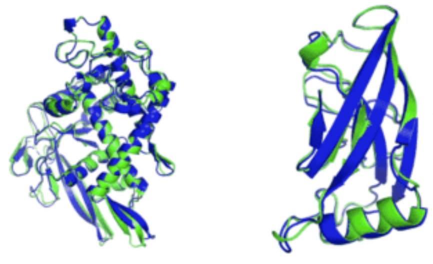


Curr. Opin. Struct. Biol. **2004**,14:70–75

The boom of AI in protein structure prediction

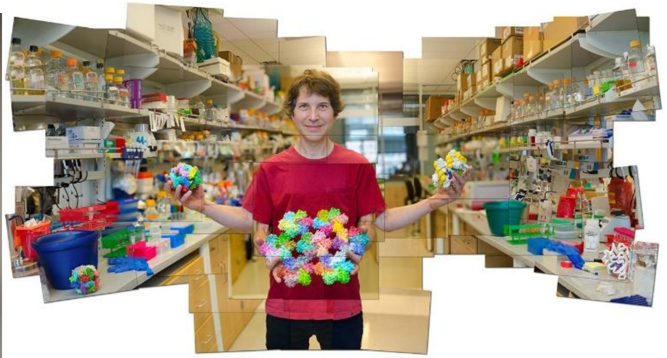
DeepMind





T1037 / 6vr4
90.7 GDT
(RNA polymerase domain)

T1049 / 6y4f
93.3 GDT
(adhesin tip)



Baker Lab

nature
https://doi.org/10.1038/s41586-021-03819-2
Accelerated Article Preview

Highly accurate protein structure prediction with AlphaFold

2021 July 16

nature
https://doi.org/10.1038/s41586-021-03828-1
Accelerated Article Preview

Highly accurate protein structure prediction for the human proteome

2021 July 22

Figure from: https://metacurate.io/

Science
RESEARCH ARTICLES
Cite as: M. Baek *et al.*, *Science* 10.1126/science.abj8754 (2021).

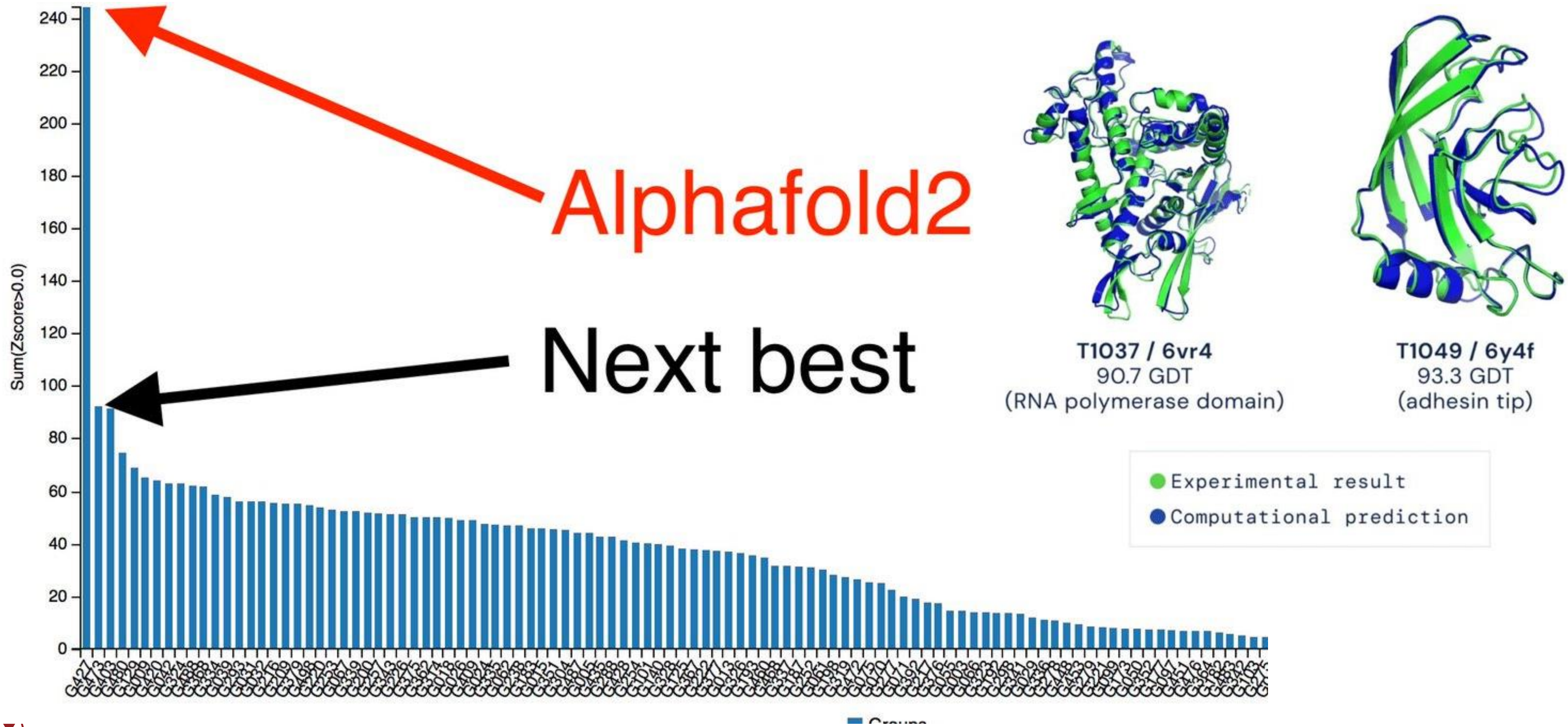
Accurate prediction of protein structures and interactions using a three-track neural network

ColabFold - Making protein folding accessible to all

Milot Mirdita,[1,*] Sergey Ovchinnikov,[2,3,*] and Martin Steinegger[4,5,*]

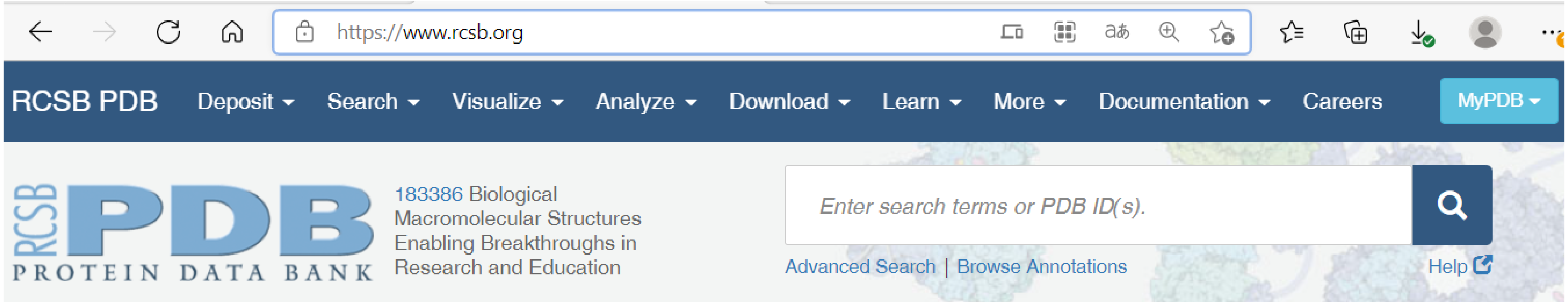
Powered by AlphaFold2 and RoseTTAFold combined with a fast multiple sequence alignment generation stage using MMseqs2.

AlphaFold: AI-based protein structure prediction



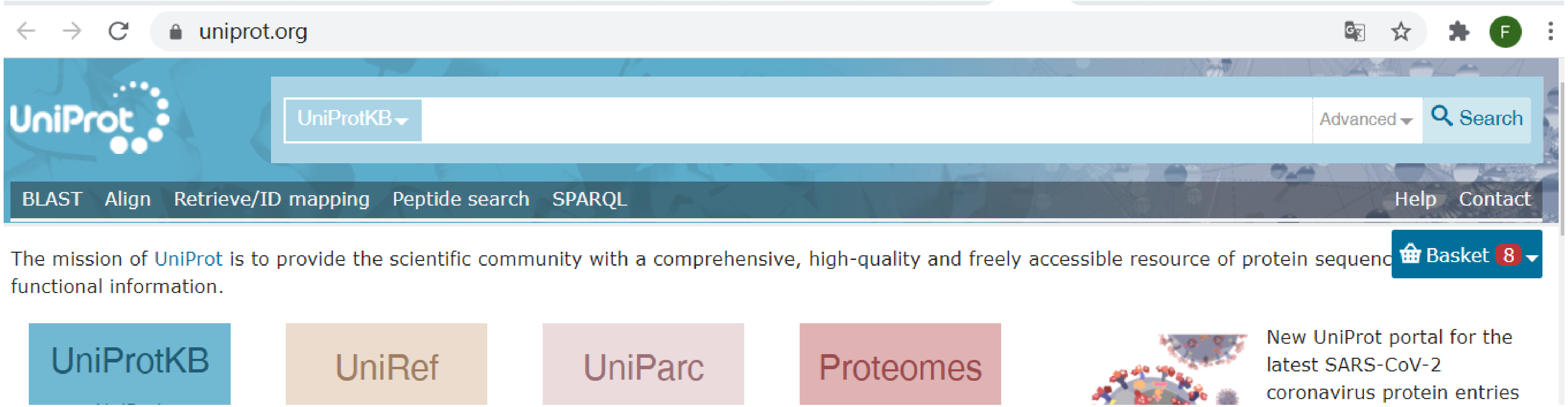
蛋白质结构数据库

https://www.rcsb.org/



蛋白序列数据库

https://www.uniprot.org/



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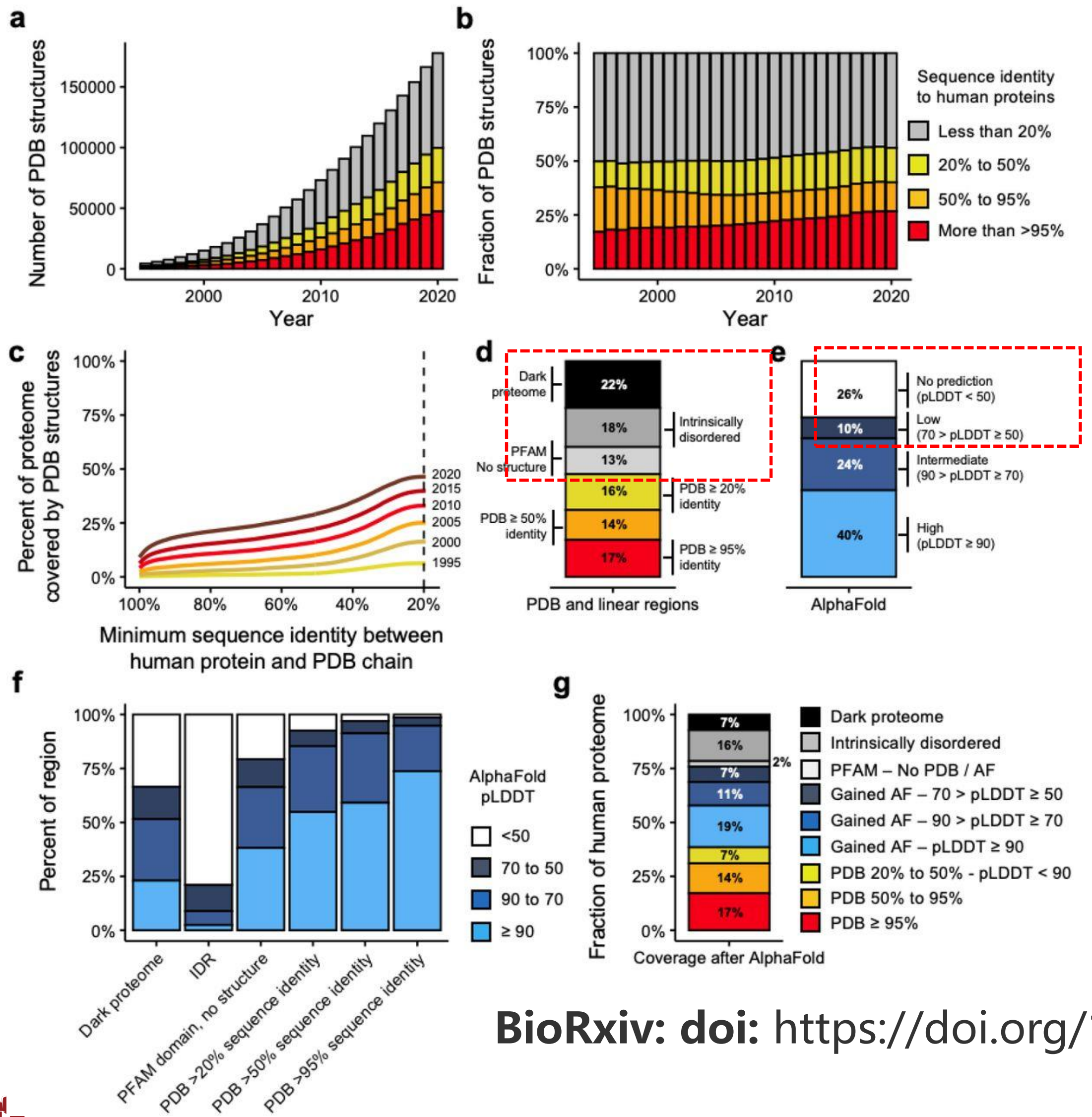
Challenges there, waiting for climbers, to be conquered…

- Number of protein structures is greatly increasing
- Human proteome is highly covered
- Large portion of the human proteome that lacks structure coverage

Current coverage of the human proteome



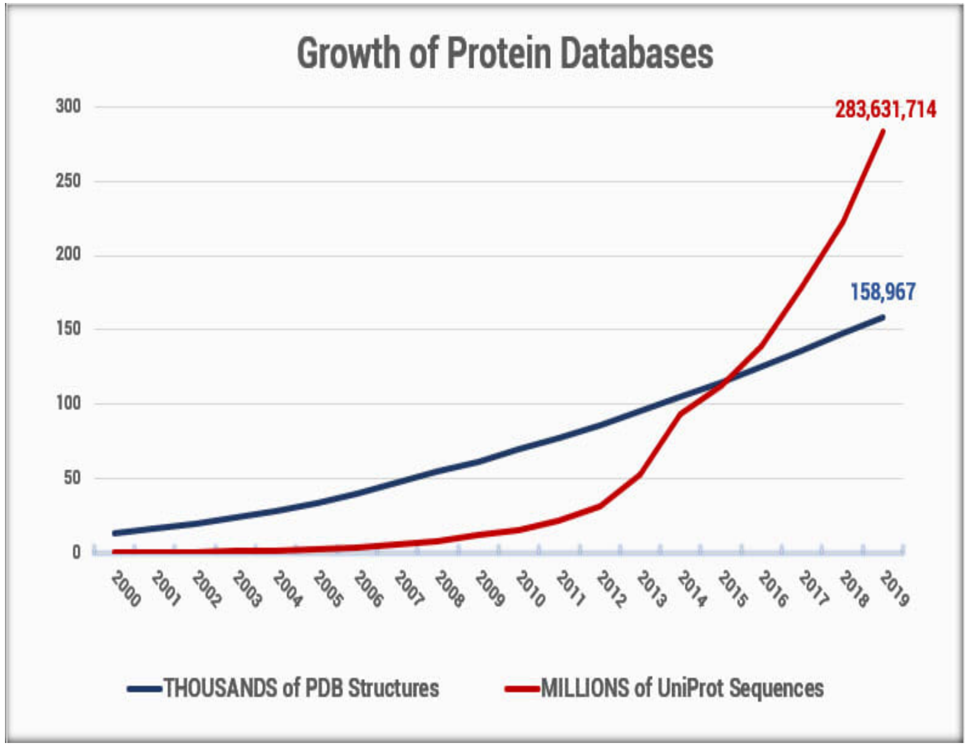
- Contribution of human proteins has also increased

Dark proteome: neither structured or predicted to be disordered

- Large portion of challengeable proteins (dark proteome, disordered proteins…)
- Large portion of low-confidential prediction
- **We have ~30% proteome to work with**

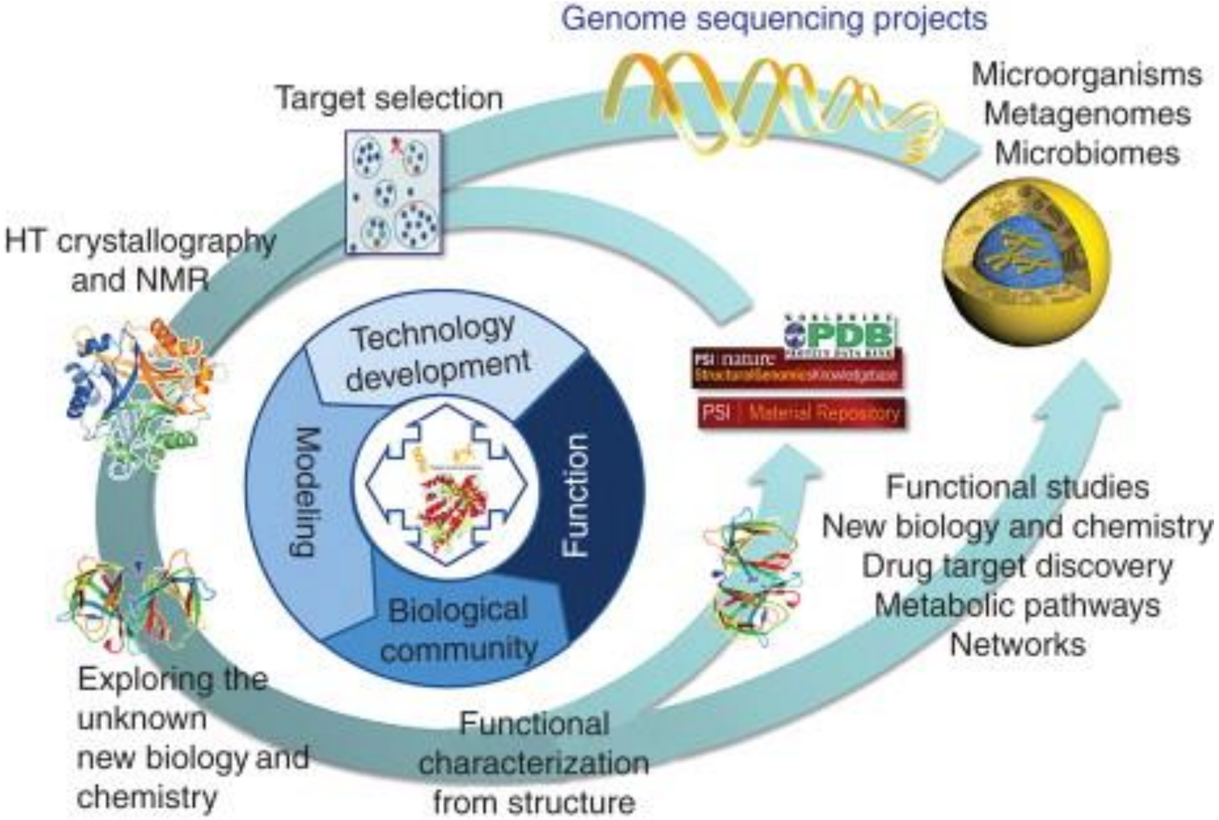
BioRxiv: doi: https://doi.org/10.1101/2021.08.03.454980

Challenges in solving protein structures



亿级

十万级

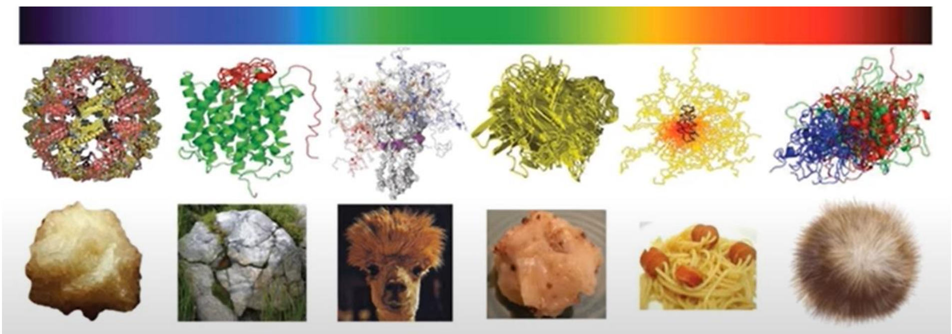


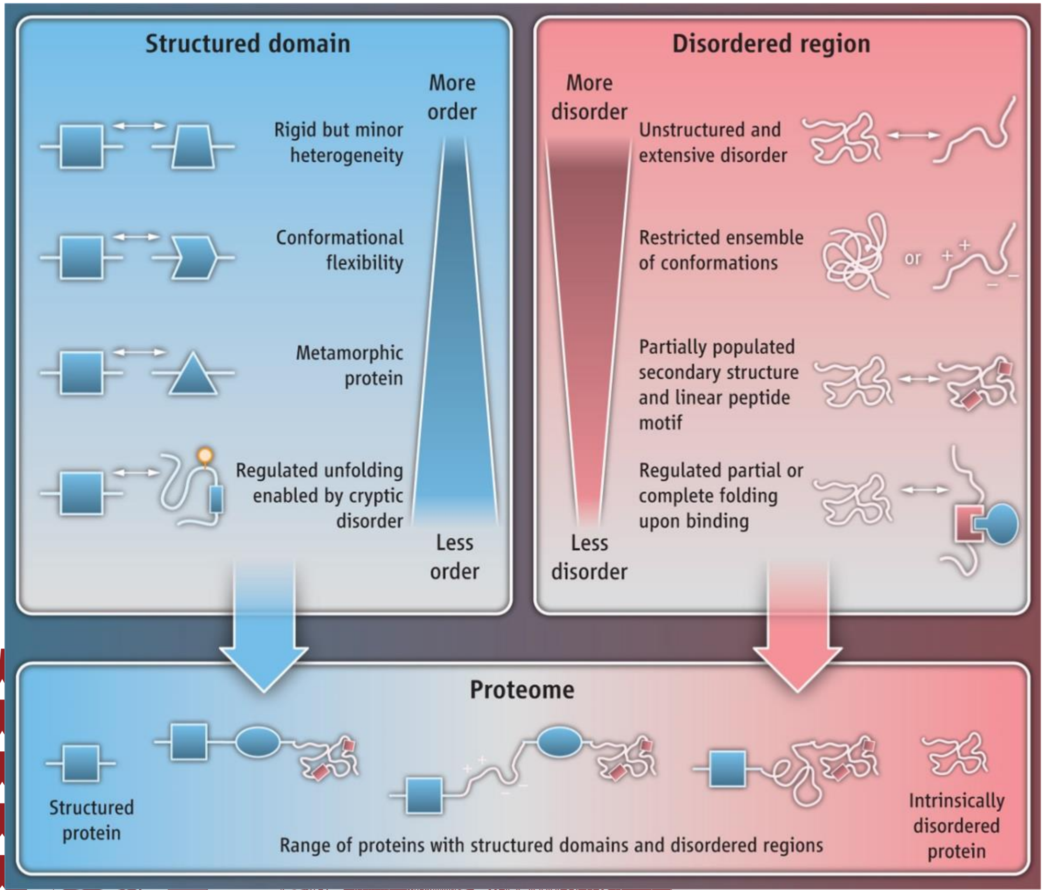
- **Growth of protein sequence and structure databases over time**

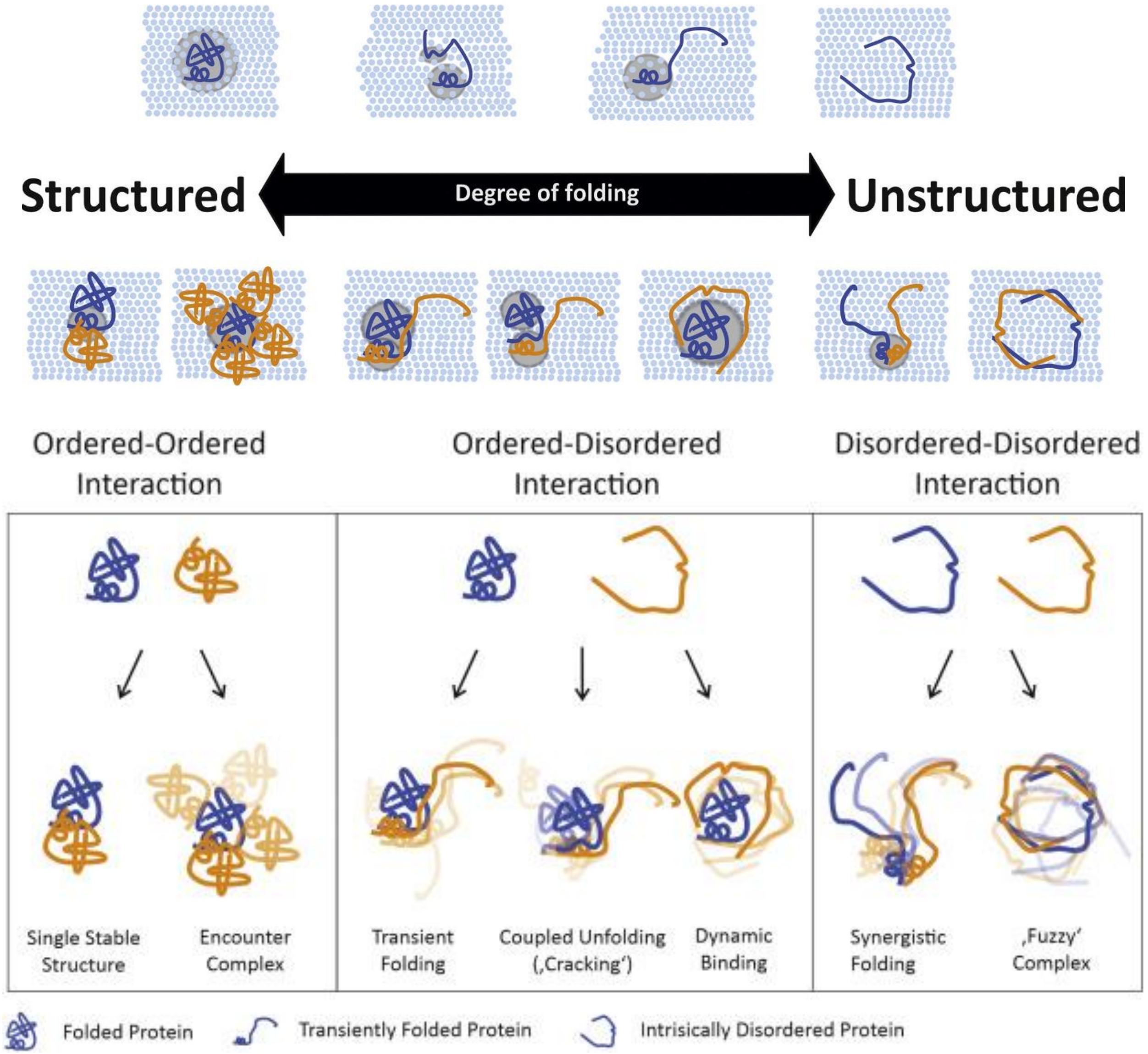
- **Genome Structure**
- **HT structural biology**

Figure from: https://www.dnastar.com

Challenges in disordered regions and proteins



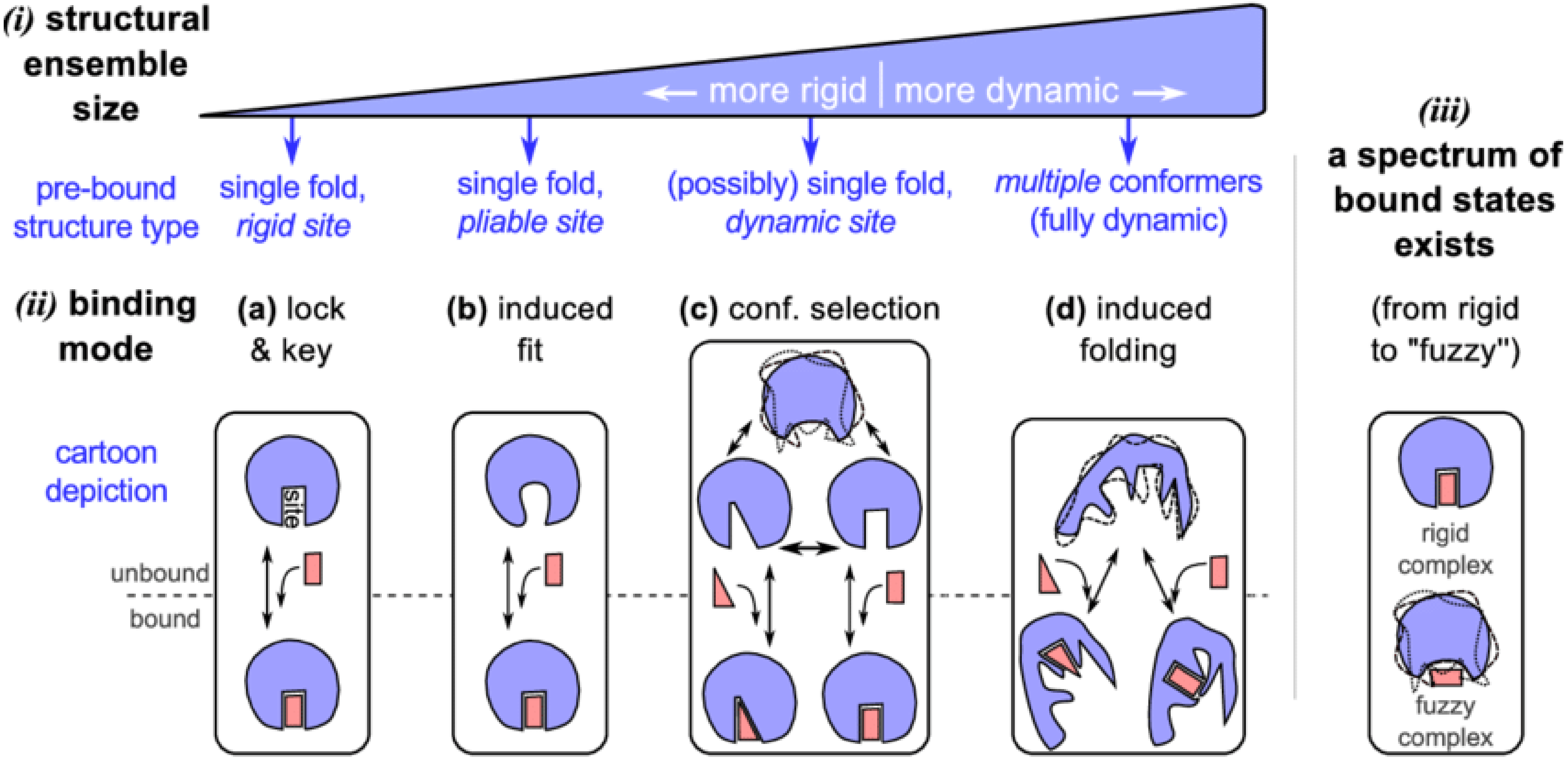




Babu et al. Science 2012;337:1460-1461

Figure (left top) from Shelly Deforte

The dynamic "new view" of protein structure and function



Proteomes, 2014, 2(1):128-153

Why structural dynamic matters?

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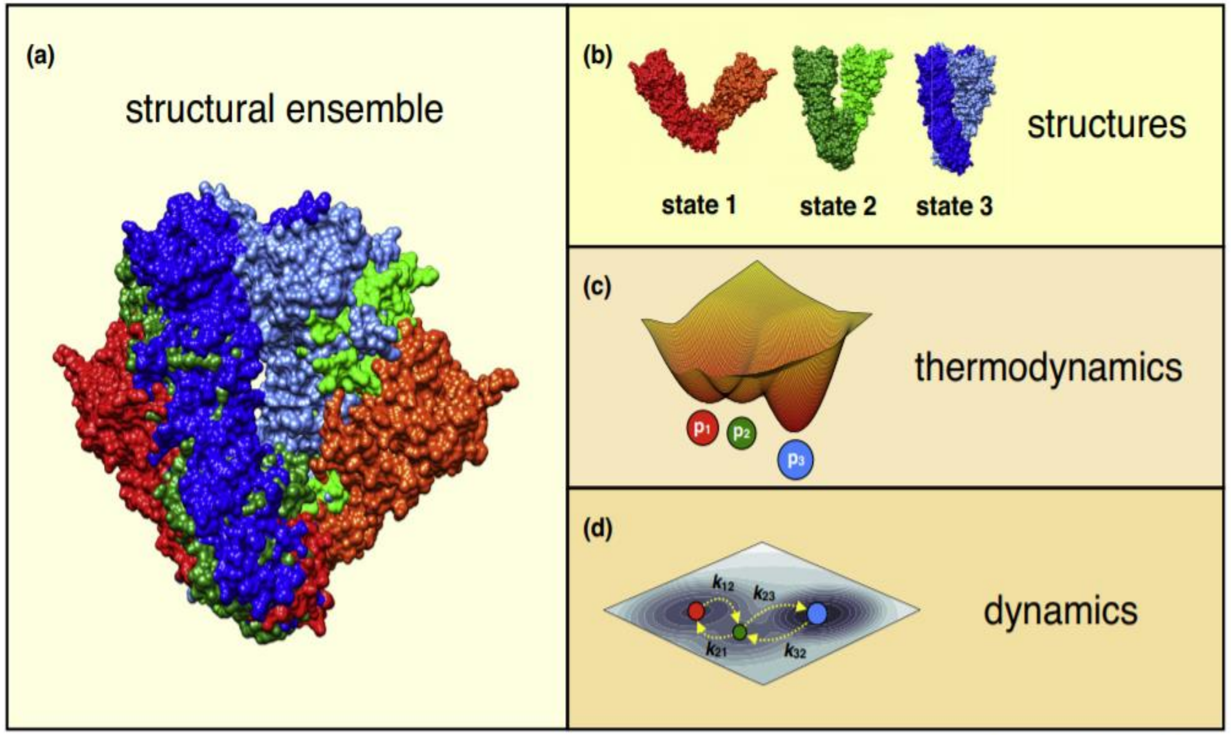
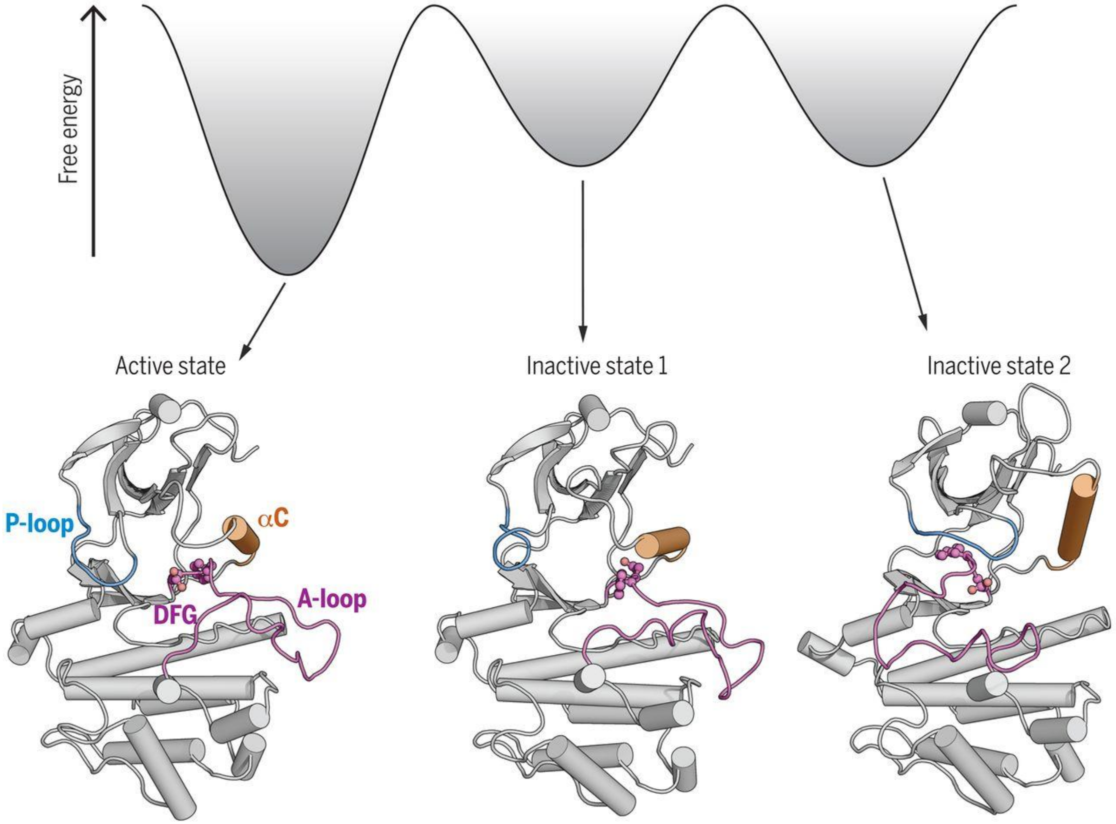
Protein structural ensembles provide unique insights to understand protein "dynamic" behavior



Current Opinion in Structural Biology

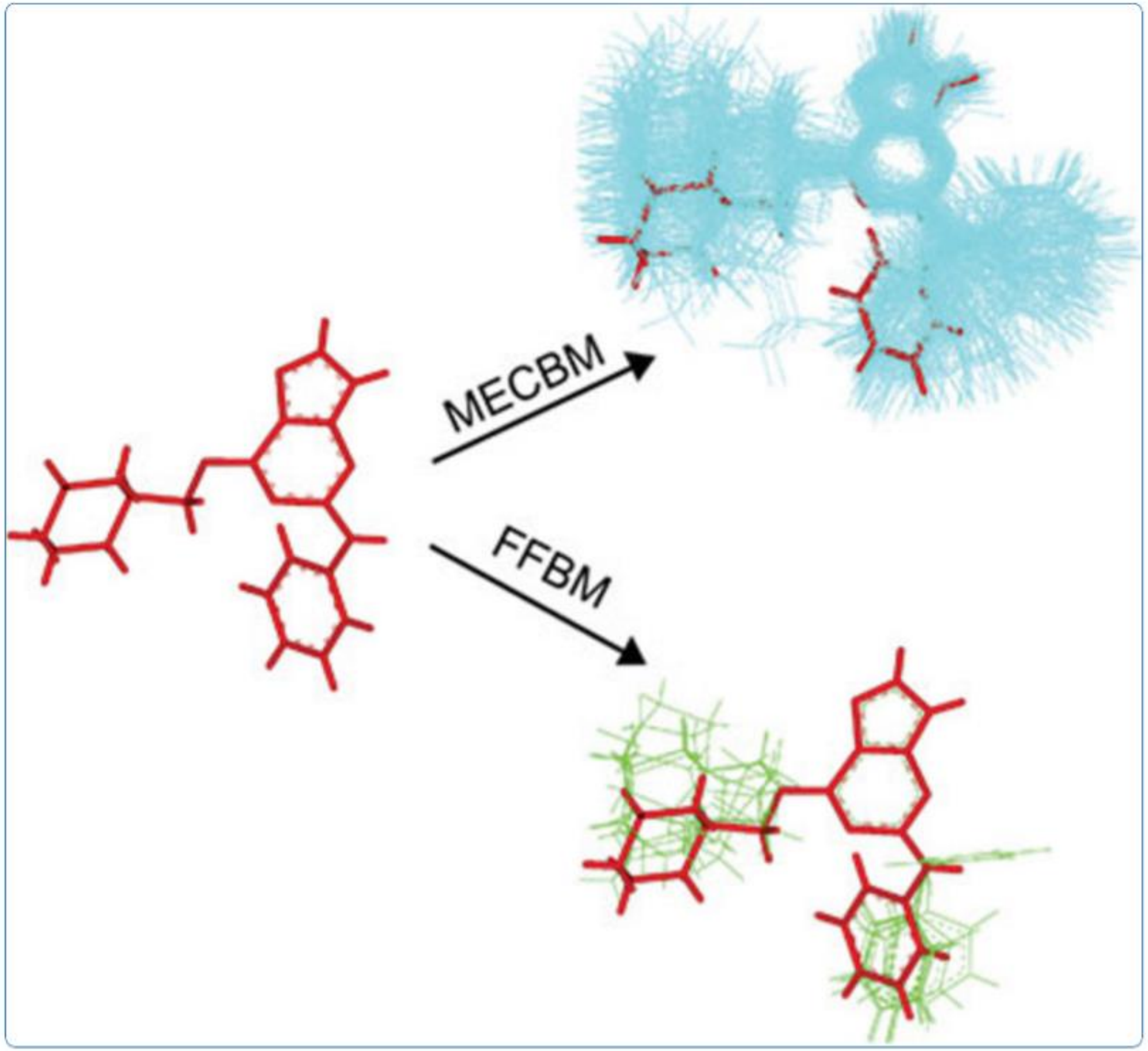
E.g. Transitions of Abl kinase between conformational states



Science, 2020, 370(6513):eabc2754

立志 成才 报国 裕民

Small molecular conformational sampling



Drug-Target Interaction

Predicting and quantifying drug-target binding

Virtual screening
- Large chem space
- Hypothesis-generating

Bioassays
- High functional relevance
- Prone to artifacts

DSF
- Fast assay development
- Indirect readout

SPR/BLI
- Kd, stoichiometry
- Immobilize protein

Ligand NMR
- Widely applicable
- Slow, material-intensive

Mass spectrometry
- Site-directed Tethering
- Native MS in development

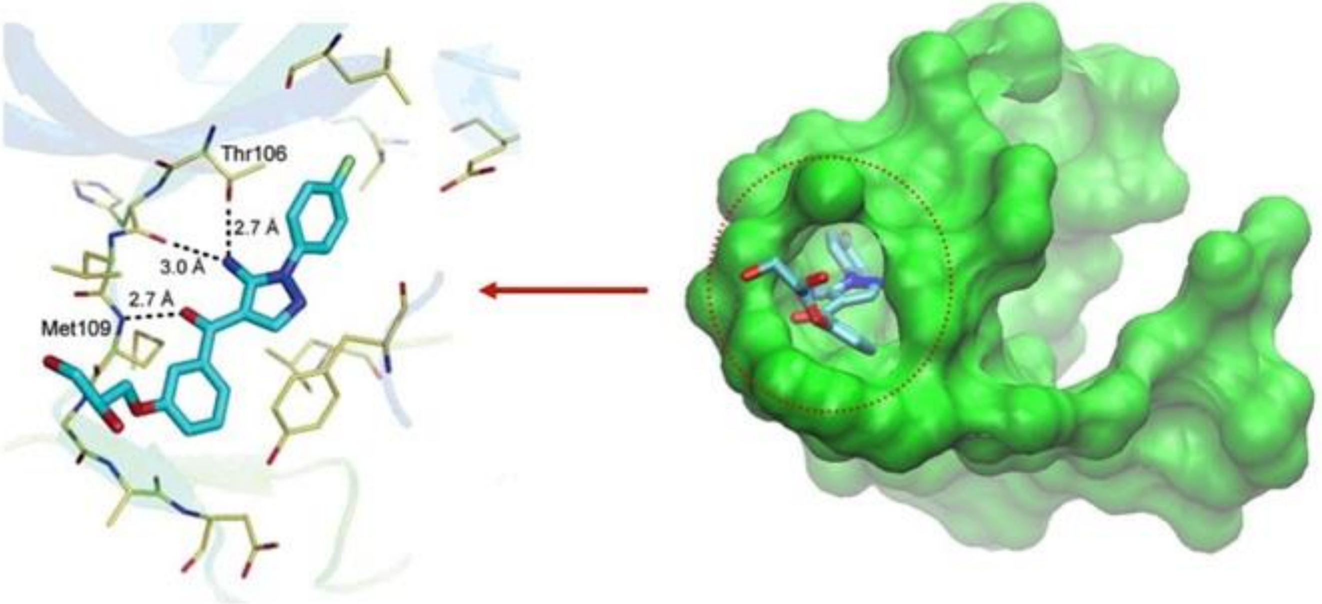
Protein NMR
- Structural information
- Size and material limitations

ITC
- Label free, Kd and stoichiometry
- Material, solubility limitations

Crystallography
- Label free, high resolution
- Prone to false -ves

Relative throughput

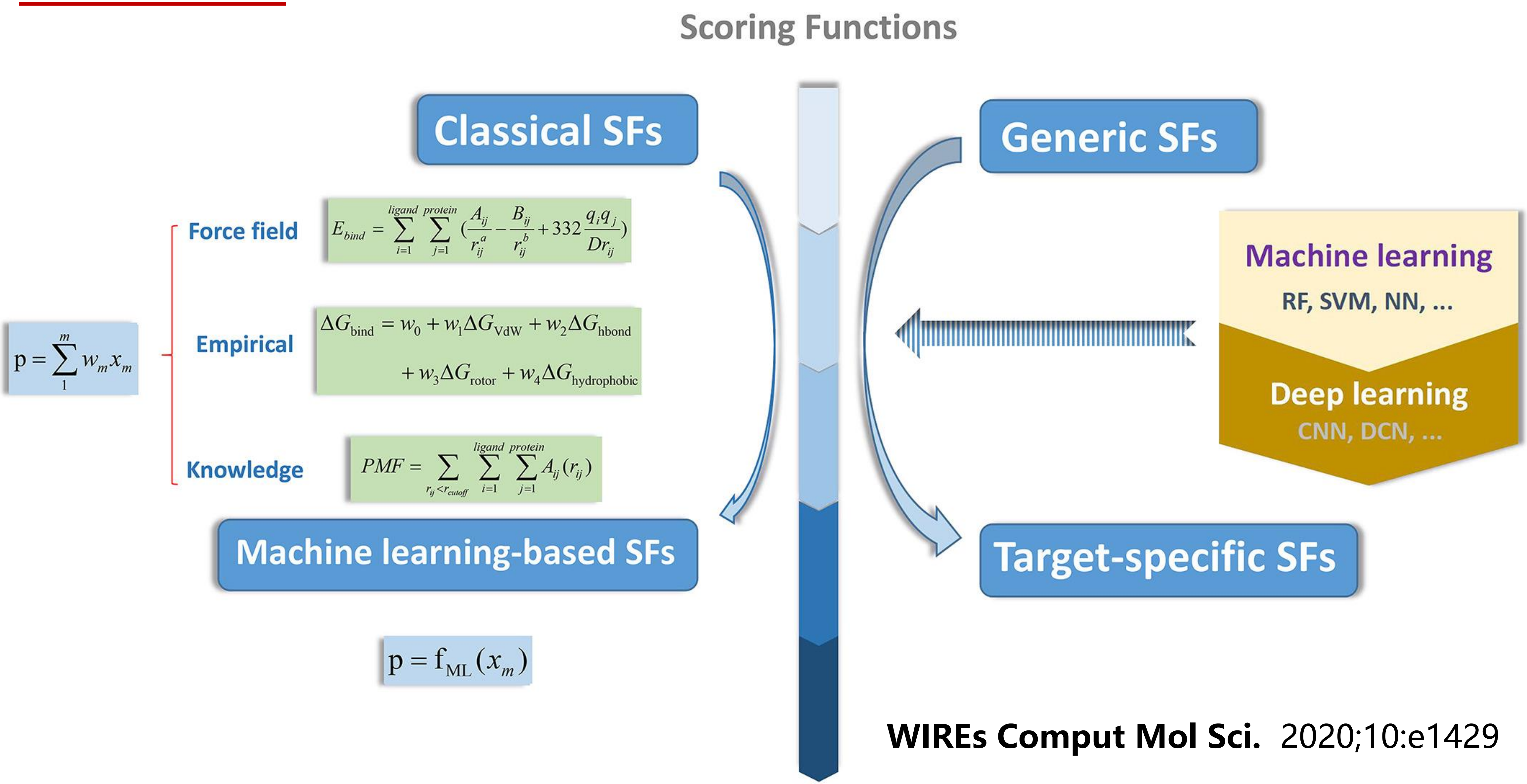
Relative information content



$$K_d = \frac{[R][L]}{[R\cdot L]}$$

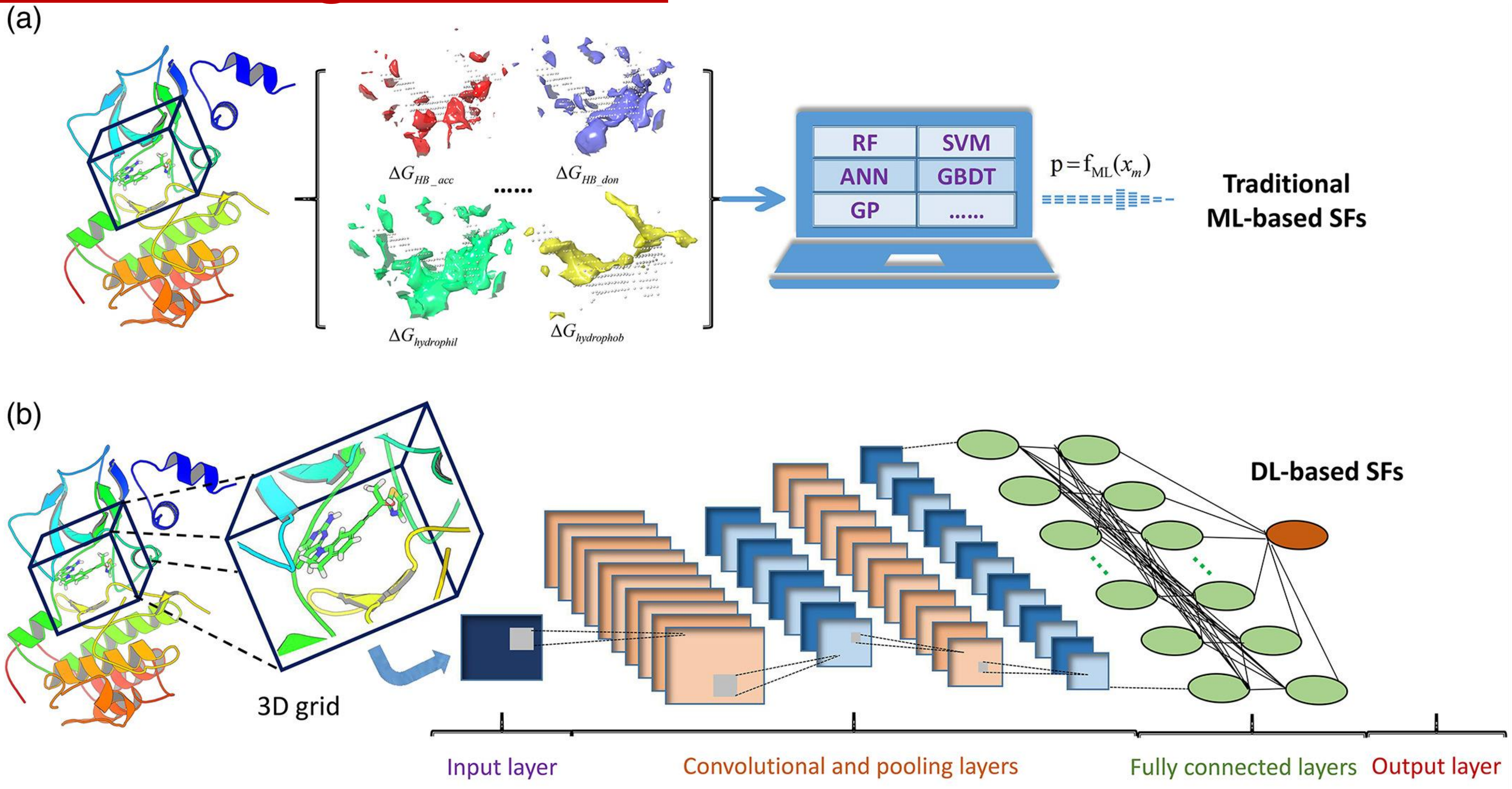
Strength of binding: R + L ←→ R•L

Estimating binding affinity by using scoring functions



WIREs Comput Mol Sci. 2020;10:e1429

Typical workflows for the development of AI-based scoring functions

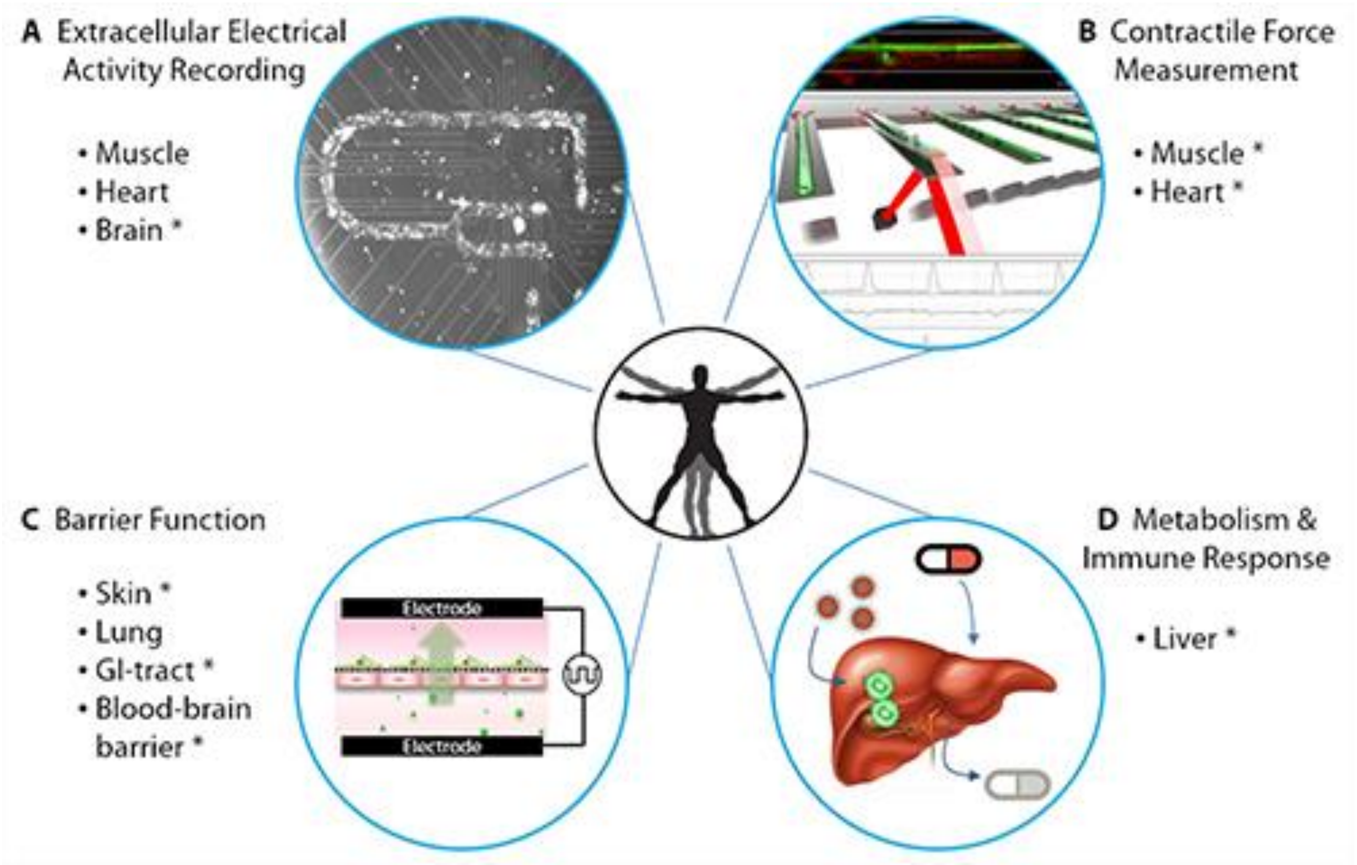


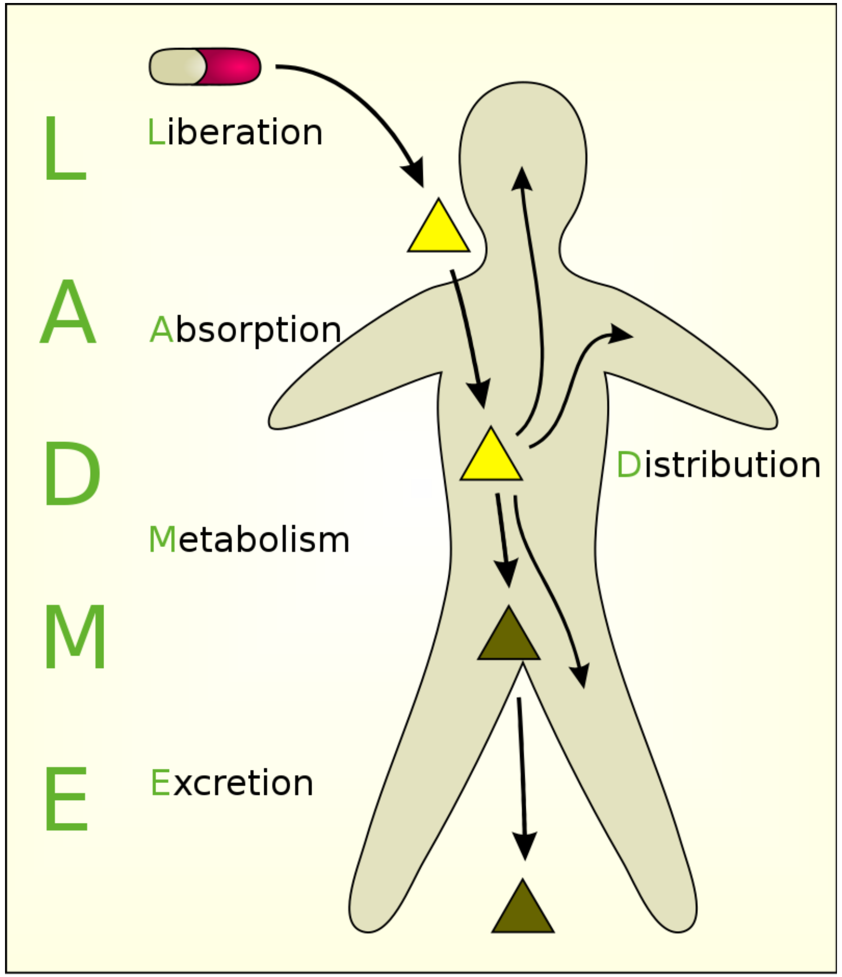
WIREs Comput Mol Sci. 2020;10:e1429

ADMET

ADMET: Turning chemicals into drugs

Optimization of Pharmacokinetics



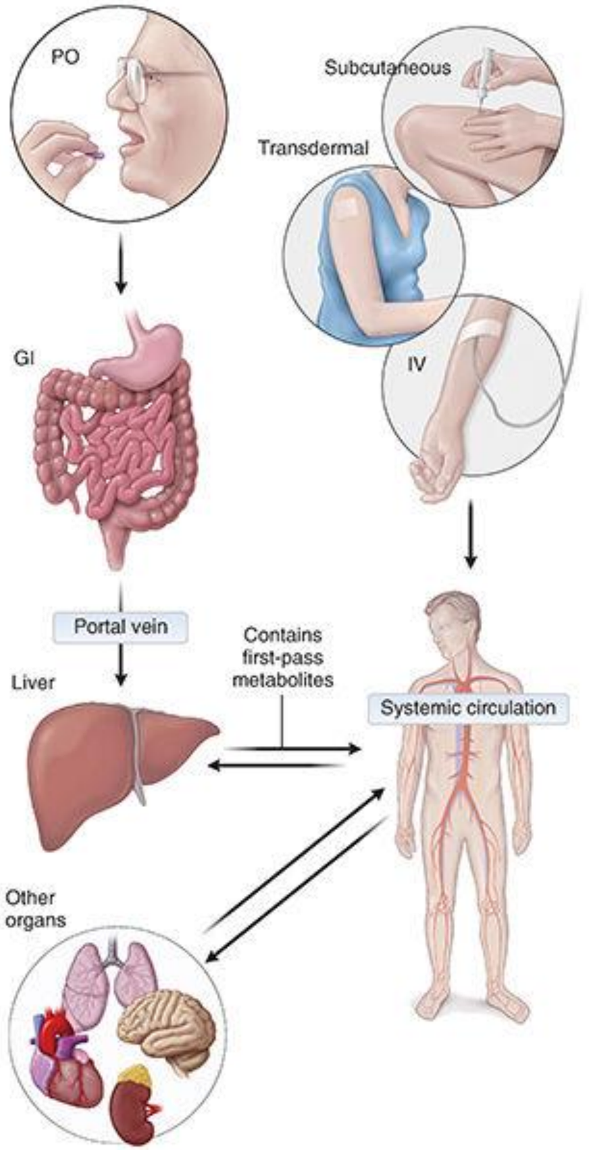


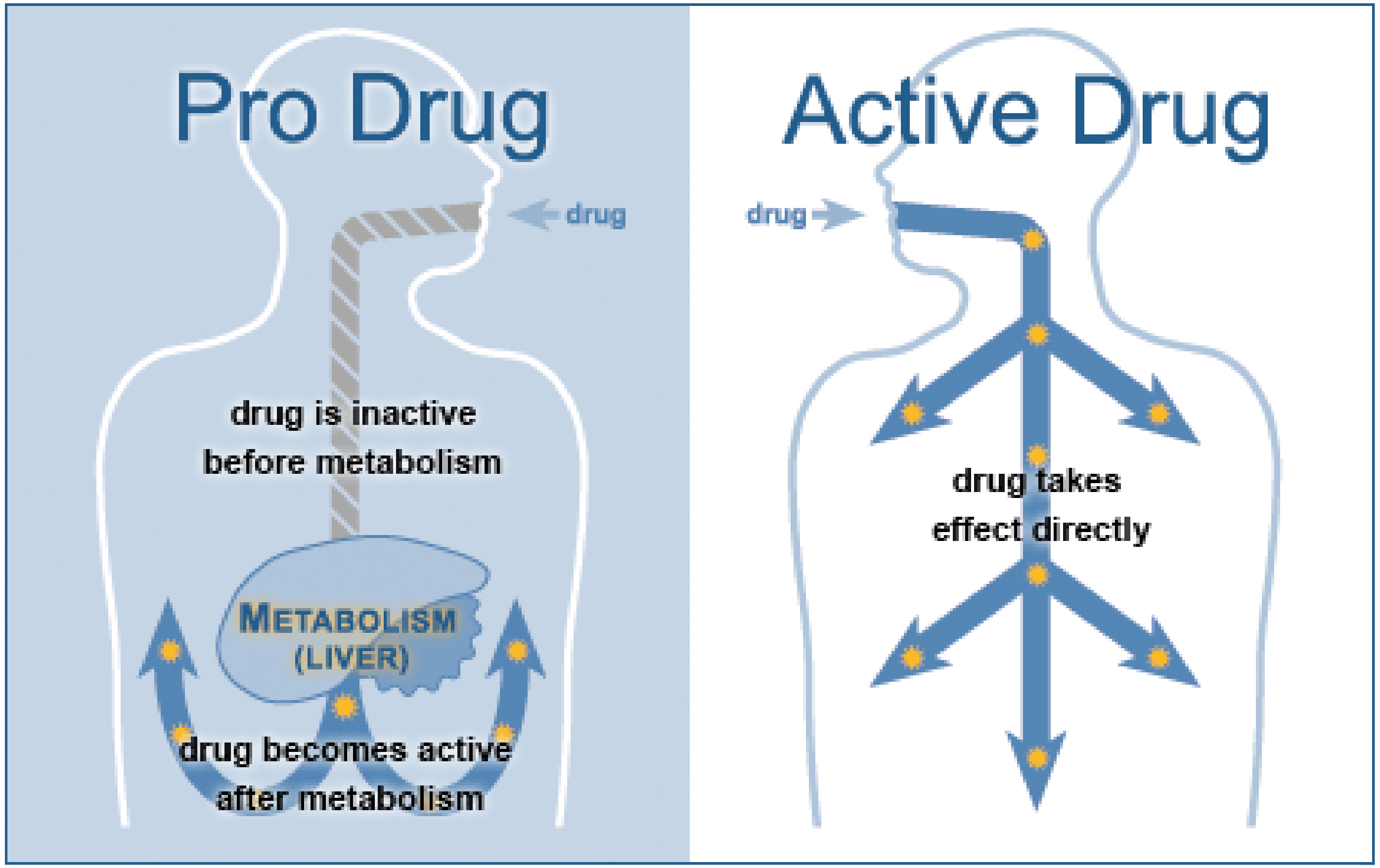
Liberation（释放）Absorption(吸收) Distribution(分布)

Metabolism（代谢） Excretion（排泄）Toxicity(毒性)

Toxicity

Drug metabolism



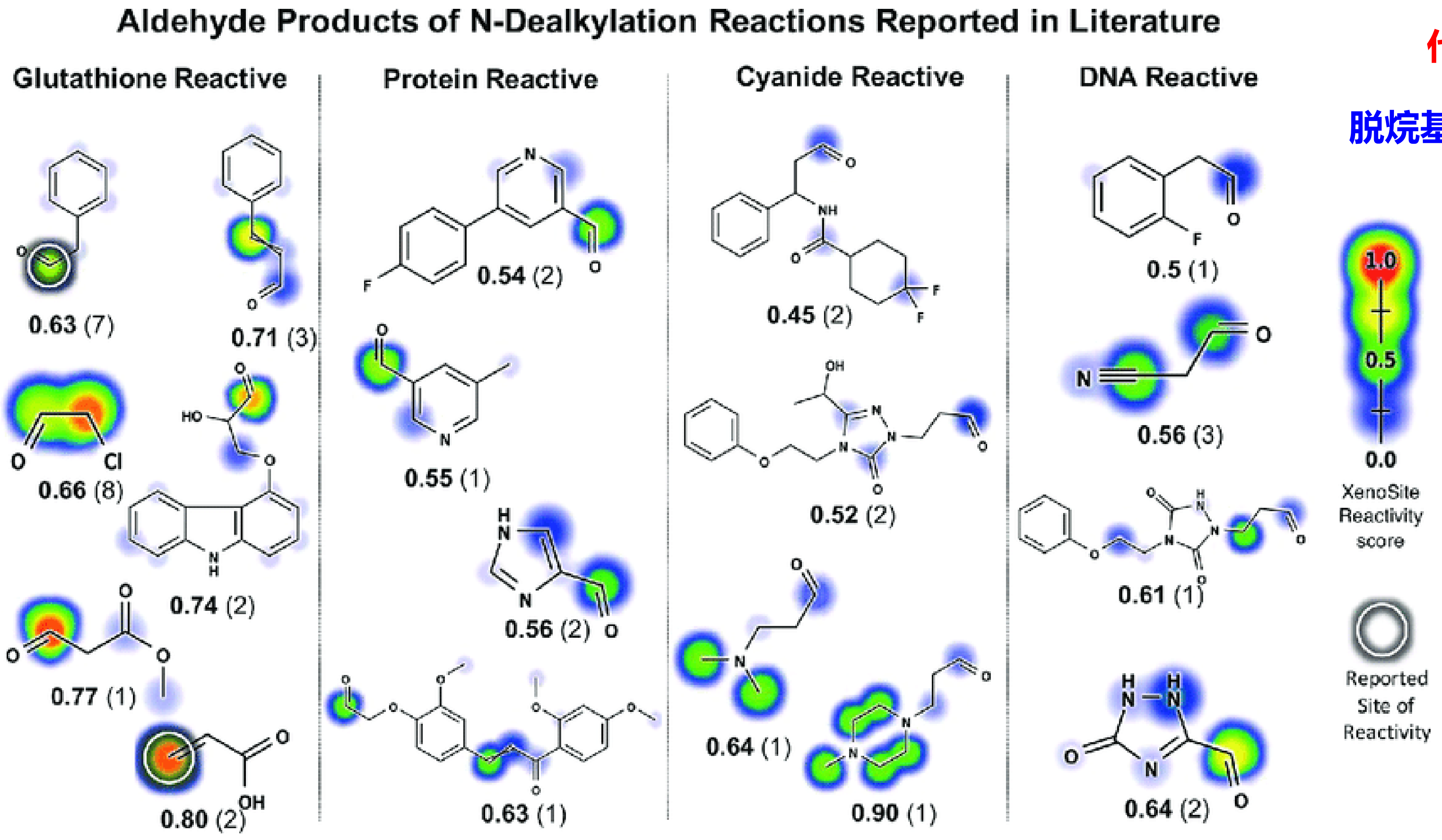


Biochemical or chemical process(es)

Prodrug → → Drug

inactive *active*

The Metabolic Rainbow: Deep Learning Phase I Metabolism in Five Colors

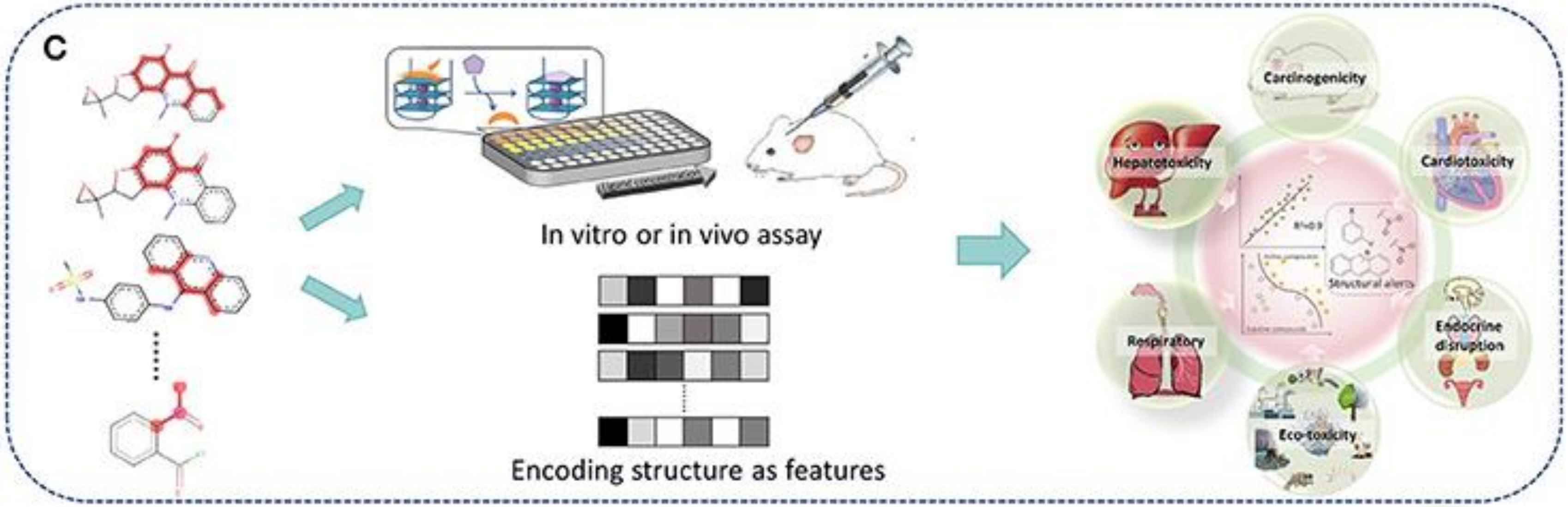


代谢彩虹

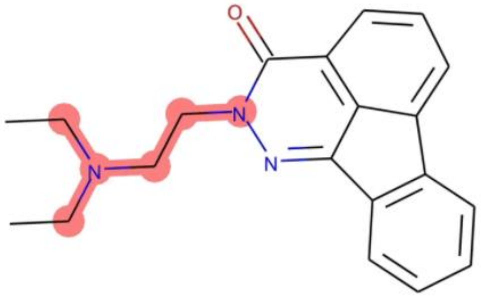
脱烷基化的醛产物

J. Chem. Inf. Model. 2020, 60, 3, 1146–1164

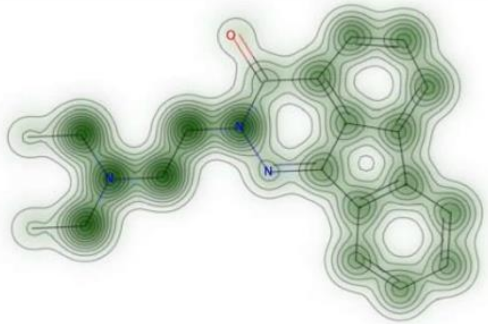
Revealing cytotoxic substructures in molecules using deep learning



Front. Chem. 2018, 6:30

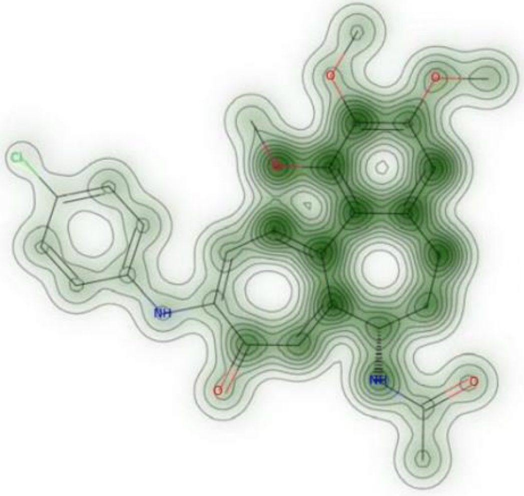


(a) Molecule 1 with bit 713 highlighted.

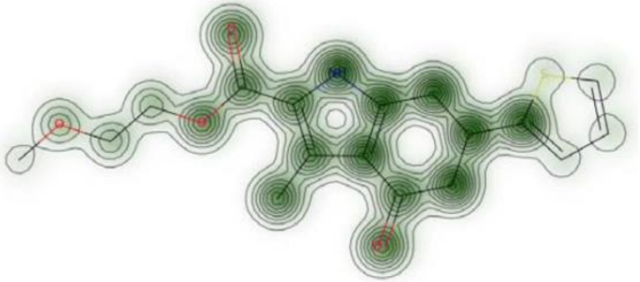


(b) Cytotoxicity map: molecule 1.

Journal of Computer-Aided Molecular Design **34**, 731–746 (2020)



(c) Cytotoxicity map: molecule 2B.



(d) Cytotoxicity map: molecule 3A.

End of Lecture 21